APPLIED ARTIFICIAL INTELLIGENCE
2026, VOL. 40, NO. 1, e2707047 (46 pages)
https://doi.org/10.1080/08839514.2026.2707047

Taylor & Francis
by informa

OPEN ACCESS Check for updates

Evolving Machine Learning in Non-Stationary Environments: A Unified Survey of Drift, Forgetting, and Adaptation

Ignacio Cabrera Martin[a], Subhaditya Mukherjee[b], Almas Baimagambetov[a], Joaquin Vanschoren[b], and Nikolaos Polatidis[a]

[a]University of Brighton, Brighton, UK; [b]Eindhoven University of Technology, Eindhoven, The Netherlands

ABSTRACT

In an era defined by rapid data evolution, traditional machine learning (ML) models often struggle to adapt to dynamic and non-stationary environments. In this work, we present evolving machine learning (EML) as a unifying paradigm for adaptive learning under distributional change, enabling continuous updating and real-time adaptation to streaming data. While prior surveys have examined individual aspects of evolving learning, such as drift detection or continual learning, there remains a lack of an integrative analysis that connects its major challenges within a coherent perspective. This survey provides a comprehensive review of EML by analyzing four interrelated challenges: data drift, concept drift, catastrophic forgetting, and skewed learning. We examine 147 recent studies, categorizing state-of-the-art approaches across supervised, unsupervised, and semi-supervised settings. Beyond taxonomic organization, we synthesize these challenges through a unified analytical perspective, highlighting their structural interdependencies and the inherent trade-off between adaptability and stability in evolving systems. Furthermore, we review evaluation protocols, benchmark datasets, and real-world applications, offering a comparative assessment of methodological strengths and limitations while identifying key research gaps and emerging opportunities for robust and scalable EML systems.

Introduction

The growth of the Internet and the increasing digitalization of modern systems have led to the continuous generation of large volumes of data from diverse sources. This has transformed domains such as finance, healthcare, cybersecurity, and autonomous systems, where data often arrive as dynamic streams rather than fixed datasets. However, traditional machine learning (ML) models are typically trained on static data and assume that future observations will follow the same distribution as the training set. In real-world environments, this assumption often does not hold, as data distributions, patterns, and relationships may evolve over time, leading to performance degradation. For example, in fraud detection, malicious behaviors constantly change as attackers adapt their strategies, making previously effective models less reliable. In healthcare, shifts in patient populations, clinical practices, or sensor measurements may reduce model accuracy and increase the risk of incorrect predictions. These challenges highlight the need for learning systems that can adapt incrementally to non-stationary environments while maintaining previously acquired knowledge. Evolving machine learning (EML) addresses this need by enabling models to update over time, respond to data drift and concept drift, reduce catastrophic forgetting, and remain robust and scalable in dynamic settings.

However, despite the success of modern ML systems, most paradigms are still based on the assumption of a static environment. Models are typically trained once on a fixed dataset obtained from a stationary distribution and then deployed with the expectation that future data will closely resemble the training data. In reality, nearly every real-world application encounters non-stationary conditions. Data distributions shift gradually or abruptly through concept drift and covariate shift, user behavior evolves, new classes appear or

CONTACT Nikolaos Polatidis N.Polatidis@brighton.ac.uk School of Architecture, Technology and Engineering, University of Brighton, Brighton BN2 4GJ, UK

© 2026 The Author(s). Published with license by Taylor & Francis Group, LLC.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

vanish, sensors degrade, and even the target task can change over time. The research community has developed numerous specialized approaches to tackle aspects of this challenge, including continual learning, lifelong learning, incremental learning, online learning, domain adaptation, transfer learning under distribution shift, and various drift detection and handling techniques. However, these efforts remain scattered across different sub-fields that employ different terminology, benchmarks, and core assumptions, which prevents the emergence of a unified theoretical foundation and broadly applicable solutions. To bring together these closely related concepts, namely the management of non-stationary data, the necessity of continuous adaptation, the structural and parametric evolution of models, and the long-term goal of autonomous improvement in open-ended environments, we propose a unifying perspective.

To provide a common perspective across these closely related research streams, we use the term EML as an umbrella designation for ML systems that continuously adapt to changing environments through online, incremental, continual, or structural learning mechanisms. We do not present EML as a new learning paradigm; rather, it serves as a unifying framework that synthesizes concepts previously studied under evolving fuzzy systems, evolving intelligent systems, continual learning, online learning, drift adaptation, and other related areas.

EML shifts the focus from models that merely react to environmental changes toward systems capable of continuously refining their knowledge, structure, and learning mechanisms throughout their operational lifetime. This transition reflects a movement from static training paradigms toward adaptive systems designed for long-term operation in dynamic environments. Although continual learning primarily addresses sequential task adaptation and catastrophic forgetting, while it encompasses a broader spectrum of challenges, including data drift, skewed learning, online adaptation, and structural model evolution. In this survey, continual learning is, therefore, treated as one methodological stream within the wider paradigm rather than as an equivalent or overarching framework.

From a business perspective, it could potentially reduce costs by automating real-time adaptation; for example, in fraud detection it may improve the identification of evolving fraudulent patterns. Scientifically, it advances continual learning, addressing non-stationary data challenges. For policymakers, it aligns with the EU AI Act (European Parliament and Council of the European Union 2024), ensuring ethical deployment through fairness and transparency. Theoretically, EML contributes to lifelong learning frameworks, while managerially, it enables scalable, autonomous systems for dynamic industries.

In contrast, online learning has gained traction as a paradigm where models are continuously updated with new data, enabling them to adapt their predictions in real-time. Traditional ML typically requires collecting all data and labels before training. Once deployed, model parameters typically remain fixed unless the system is retrained from scratch, an approach poorly suited to real-world, evolving scenarios. EML extends online learning by incorporating methods that specifically address dynamic data distributions, drift, and adaptation challenges, making it increasingly relevant in industry and research.

By incrementally incorporating new data, online learning reinforces predictive power and often improves model performance. The exponential rise of the Internet of Things (IoT) has further accelerated the need for EML, as connected devices now generate massive real-time data streams that must be processed with minimal delay. This survey reviews these techniques in detail and analyzes their suitability for different categories of tasks based on specific attributes.

Beyond technical advances, the growing role of AI in high-stakes applications such as fraud detection and predictive maintenance has driven policymakers to engage with AI experts worldwide. Their goal is to establish frameworks that ensure AI development aligns with ethical principles. For example, the EU AI Act (European Parliament and Council of the European Union 2024), the UK Parliament's Research briefing (Bhatnagar and Gajjar 2025), and the Guidelines for Trustworthy AI (High-Level Expert Group on Artificial Intelligence 2019) emphasize requirements such as human oversight, robustness, privacy and data governance, transparency, fairness, societal well-being, and accountability. This survey considers these factors while discussing methods and their deployment.

As with any approach, online learning and EML face significant challenges, including data and concept drift, catastrophic forgetting, and skewed learning. While prior surveys address individual aspects of evolving learning, such as drift detection or continual learning, there is no unified analysis integrating all four core challenges under a single paradigm (Johnson and Khoshgoftaar 2019; Lu et al. 2019). However, a holistic analysis of EML's core challenges, with an emphasis on

efficiency and ethics, is still lacking. This survey aims to establish a unified taxonomy and analytical paradigm that connects these challenges to methodological strategies and evaluation criteria, as well as the challenges that remain open. Unlike existing surveys on drift detection, class imbalance, or continual learning, this work provides a holistic taxonomy across these challenges, integrates ethical and regulatory aspects, and synthesizes case studies. To the best of our knowledge, this survey provides one of the most comprehensive reviews of EML as an overarching paradigm, integrating data drift, concept drift, catastrophic forgetting, and skewed learning within a unified framework, where our findings reveal that adaptive ensembles and meta-learning strategies consistently outperform static architectures under combined data and concept drift, yet lack standardized benchmarks for cross-domain validation. Finally, a preprint of the survey can be found in Cabrera Martin et al. (2025).

The main contributions of this survey are as follows:

- We conceptualize EML as a unifying survey framework that integrates traditionally fragmented research areas, including continual learning, drift detection, skewed learning, and adaptive model evolution.
- We derive an integrative taxonomy that connects detection, mitigation, and adaptation strategies across these challenges, highlighting their structural interdependencies.
- We provide a comprehensive synthesis of recent methods, datasets, and evaluation protocols, identifying theoretical gaps and open research directions necessary for sustained learning under non-stationarity.

The remainder of this paper is organized as follows. Setion "Survey methodology" presents the survey methodology, including the databases searched, search strategy, time frame, inclusion and exclusion criteria, and screening process. Setion "Background of evolving machine learning" provides the background of EML, outlining its definition, key characteristics, relationship with traditional machine learning, and evaluation considerations. Setion "EML challenges" examines the main challenges addressed in this survey, namely data drift, concept drift, catastrophic forgetting, and skewed learning. Setion "Comparative analysis and case studies" presents a comparative analysis of the reviewed methods and discusses representative real-world and benchmark-based applications. Setion "Future research directions" identifies open challenges and future research directions for robust, efficient, and trustworthy EML systems. Finally, Setion "Conclusion" concludes the paper by summarizing the main findings and implications of the survey.

Survey methodology

Databases searched

The literature search was conducted across several major scientific indexing and digital library platforms that provide broad coverage of artificial intelligence, machine learning, data mining, and computational intelligence research. The main databases searched were Scopus, IEEE Xplore, ACM Digital Library, SpringerLink, and Web of Science. These sources were selected because they collectively index a wide range of peer-reviewed journal articles, conference proceedings, book chapters, and survey papers relevant to EVL, continual learning, adaptive systems, and learning under non-stationary data distributions.

Search strategy

The search combined keywords such as "evolving machine learning," "concept drift," "catastrophic forgetting," "online learning," "adaptive neural networks," and "class imbalance streams." Related terms, including "continual learning," "incremental learning," "data streams," "drift detection," and "imbalanced learning," were also considered to capture relevant studies using different terminology. Boolean operators such as AND and OR were used to combine concepts, broaden the search scope, and refine the results according to the main themes of the survey.

Time frame

The review primarily focused on studies published between 2018 and 2024, as this period reflects the recent growth of research on EML, continual adaptation, non-stationary learning environments, and stream-based learning. To provide sufficient background and methodological continuity, relevant works published after 2015 were also considered, especially where they contributed to the development of current approaches. Earlier publications were included selectively when they introduced foundational concepts, algorithms, or evaluation practices that continue to influence the field. Examples include seminal work on drift detection, catastrophic forgetting, and imbalanced learning, such as ADWIN for adaptive windowing and drift detection, EWC for continual learning and catastrophic forgetting, and SMOTE for synthetic minority oversampling. This approach allowed the survey to balance recency with recognition of foundational contributions.

Inclusion criteria

Peer-reviewed journal or conference publications, papers proposing new algorithms, frameworks, or benchmarks for EML and surveys or comparative studies addressing adaptation in non-stationary environments.

Exclusion criteria

Studies were excluded if they were not written in English, were duplicates, lacked sufficient methodological detail, or did not provide empirical validation or meaningful theoretical insight. Papers were also excluded where adaptation, drift, continual learning, or imbalance handling was only mentioned superficially and did not form a substantive part of the contribution while non-peer-reviewed material was excluded.

Screening and selection

The initial queries returned several hundred papers. After applying the criteria above, we retained a set of the most relevant works that span the four core challenges of EML: data drift, concept drift, catastrophic forgetting, and skewed learning. These form the basis of the taxonomy and synthesis presented in this survey.

Background of evolving machine learning

The notion of evolving learning systems has a long history in the literature, particularly within evolving fuzzy systems, evolving intelligent systems, and evolving connectionist systems (Angelov and Zhou 2006; Angelov, Filev, and Kasabov, 2010; Lughofer, 2011; Angelov, 2023; Kasabov, 2007). Earlier works established principles such as incremental knowledge acquisition, online adaptation, structural evolution, and learning under non-stationary environments. In this survey, we adopt and extend these ideas beyond fuzzy-system implementations and use the term EML to encompass a broader range of adaptive machine learning approaches, including neural, probabilistic, ensemble-based, and continual-learning methods. This section provides the background of EML by covering its definition and the key characteristics, its relationship with traditional machine learning, along with its differences, and the unique aspects of evaluation in these kinds of systems.

Definition and characteristics

EML refers to ML systems that are designed to adapt continuously to changing data distributions and evolving environments through incremental or online learning mechanisms (Angelov and Zhou 2006; Angelov, Filev, and Kasabov, 2010; Lughofer, 2011; Angelov, 2023; Kasabov, 2007). Unlike traditional ML approaches that are trained on a fixed dataset and subsequently deployed, EML systems incorporate adaptation strategies that allow models to update their behavior as new data become available. These

strategies can be applied at different stages of the ML pipeline, including the preprocessing stage (feature selection and resampling methods), model learning (e.g., parameter adaptation, evolving neural architectures, and ensemble updating), and postprocessing (e.g., rule optimization, pruning, and decision aggregation). Such parameter and structural adaptation mechanisms are also central to earlier evolving fuzzy, neuro-fuzzy, and evolving connectionist systems (Angelov and Zhou 2006; Lughofer, 2011; Kasabov, 2007). The core characteristics of EML systems contribute significantly to their effectiveness and suitability for dynamic environments. Thus, this ongoing adaptation allows EML systems to quickly respond to emerging patterns, thereby maintaining performance levels without substantial degradation.

Several characteristics commonly appear across the EML literature. First, incremental learning enables models to assimilate new information while attempting to preserve previously acquired knowledge, a principle already central to evolving fuzzy systems and evolving connectionist systems (Angelov, 2023; Lughofer, 2011; Kasabov, 2007). This characteristic is particularly important in non-stationary environments, where retraining a model from scratch may be impractical or computationally expensive. Methods such as iCaRL, EWC, and OWA illustrate different strategies for balancing adaptation to new information with the preservation of previously learned representations.

Second, many EML approaches incorporate mechanisms for monitoring changes in the incoming data stream, which is consistent with the data-stream orientation of earlier evolving intelligent systems (Angelov and Zhou 2006; Angelov, Filev, and Kasabov, 2010). Drift-detection methods such as Adaptive-SPLL and DetectA continuously analyze statistical properties of the data and trigger adaptation procedures when significant distributional changes are detected. These mechanisms are intended to support model maintenance in dynamic environments, although the extent of automation varies considerably across methods and application domains.

Third, real-time or near-real-time operation is a common design objective in EML systems. Online and streaming-learning approaches such as SEOA and OWSELM are specifically designed to process continuously arriving data while updating predictive models incrementally. Such approaches are particularly relevant in domains where delayed adaptation may lead to performance degradation.

Finally, scalability remains an important consideration in evolving systems. Several approaches reviewed in this survey, including adaptive ensembles and dynamically expandable architectures, seek to accommodate increasing data volumes and evolving task requirements while maintaining computational feasibility. However, scalability remains an open challenge, particularly in large-scale and resource-constrained environments.

While EML draws inspiration from evolving fuzzy systems, evolving intelligent systems, and evolving connectionist systems, it is broader in scope. Traditional evolving-system research primarily focused on adaptive fuzzy-rule-based and hybrid intelligent systems, whereas EML encompasses the wider ML ecosystem, including deep neural networks, continual-learning methods, adaptive ensembles, online optimization techniques, and foundation-model adaptation. Therefore, EML should be viewed as a contemporary umbrella framework rather than a replacement for earlier evolving-system concepts.

Relationship with traditional machine learning

ML models aim to provide predictive capabilities and automate repetitive tasks involving data analysis and decision-making, it is relatively normal to say that traditional ML and EML share the foundational principles and methodologies. In some cases, EML can be viewed as an extension and enhancement of traditional ML, where they are often built upon foundational algorithms such as Decision Trees, Support Vector Machine (SVM), or neural networks, and simply address the limitations that traditional ML poses with dynamic and non-stationary data streams. Both methodologies (ML and EML) share significant similarities, primarily as they are developed for the discovery of meaningful patterns within data sets. They strive for robustness by optimizing predictive accuracy and simultaneously minimizing prediction errors, thereby ensuring reliable outcomes. Furthermore, these approaches significantly contribute to enhancing decision-making capabilities by effectively integrating and analyzing the available data. Central to both methodologies are tasks involving classification and regression, highlighting their versatility and broad applicability across various analytical scenarios.

Both methodologies support supervised and unsupervised learning techniques, reflecting their comprehensive nature and flexibility in addressing diverse analytical problems. Additionally, they utilize sophisticated feature engineering techniques such as normalization, encoding of categorical features, and dimensionality reduction, all essential for improving the overall efficiency and accuracy of data models. Reliance on optimization methods such as gradient descent or Adam, alongside regularization techniques like dropout or batch normalization, underscores their commitment to model stability and accuracy. Lastly, both methodologies employ crucial evaluation metrics, including Accuracy, Precision, Recall, F1-score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and ROC curves, to rigorously assess and ensure the effectiveness of the developed models.

On the other hand, the primary distinction between EML and traditional ML lies in adaptability. EML continuously adjusts as data evolves, dynamically responding to changes. In contrast, traditional ML systems exhibit limited adaptability, typically requiring periodic updates to remain effective, while feature extraction and selection methods remain critical within the EML context. However, these methods have evolved significantly to operate dynamically, continuously adapting to incremental data streams. This ongoing adjustment capability ensures that EML systems remain relevant and effective even as data characteristics shift.

Regarding evaluation metrics, EML introduces specialized measures tailored to the unique challenges of dynamic data streams. These new metrics specifically quantify aspects such as the degree of data change, including indicators like forgetting rate and adaptability scores. These metrics provide essential insights into how effectively an EML system adapts over time.

Another advantage of EML systems is their reduced maintenance requirement. Due to their inherent adaptive nature, these systems can autonomously adjust, minimizing the need for manual interventions and updates. Consequently, EML systems offer efficiency benefits through lower maintenance efforts compared to traditional ML.

Moreover, robustness is another notable strength of EML systems, as they are explicitly designed to detect and adapt to data drifts effectively. This feature significantly enhances their reliability in environments characterized by frequent or unpredictable data changes.

Finally, catastrophic forgetting, a phenomenon where previously learned information is lost when new data is incorporated, is generally not a concern in traditional ML due to its static data approach. However, managing catastrophic forgetting is a critical aspect of EML, as systems must retain previously acquired knowledge while adapting to new information streams.

Evaluation in evolving machine learning

Since EML deals with changing environments, we need dynamic metrics that assess performance over time. In this kind of scenario, traditional static metrics are not as effective as they do not consider the different challenges associated with continuous learning, and it is required to use more specialized evaluation metrics that capture the model's accuracy and adaptability over time, and this is the case of dynamic techniques and metrics.

For consistency throughout this subsection, let X denote the covariate (feature) space and Y the class-label space. Lowercase symbols $x \in X$ and $y \in Y$ denote individual feature values and class labels, respectively. The symbols $P_t(\cdot)$ and $P_u(\cdot)$ represent probability distributions at time periods t and u.

Meta-learning and transfer-based approaches

When referring to techniques, the Prequential Evaluation, specifically the Interleaved Test-Then-Train approach, is a sequential evaluation technique that involves alternating between testing and training phases for each data instance. This method assumes that every individual instance initially serves to evaluate the model's accuracy and subsequently updates the model through training, aiming to progressively enhance model accuracy and efficiently utilize available data. The prequential error, a crucial component of this evaluation, is computed cumulatively through a loss function that compares predicted values to actual observed outcomes, formally represented in Equation 1, where it takes a sequence of observations and represents the loss function used for the type of problem, taking as the prediction made by the model before seeing the true label:

$$Prequential\ Error = \sum_{t=1}^{n} f(\hat{y}_t, y_t) \tag{1}$$

There are three primary prequential evaluation techniques commonly implemented across various approaches, each addressing specific scenarios and data characteristics: Landmark Window (Interleaved Test-then-Train), Sliding Windowing, and Forgetting Mechanisms. Each of these methods offers a unique strategy for managing data streams and dealing with evolving data distributions, reflecting different perspectives on how past data relevance should diminish over time. Additionally, the Controlled Permutations evaluation approach addresses the challenge of accurately assessing adaptive learning algorithms. This method reduces the masking of adaptive properties – common when averaging performances across various algorithms – by running multiple tests on randomly permuted versions of data streams. By preserving different distributions within these permutations, this technique effectively measures model robustness and adaptability, particularly beneficial in scenarios involving sudden concept drifts. Controlled permutations create diverse test sets, enabling the evaluation of model volatility, robustness, and parameter optimization, thus minimizing overfitting risks associated with sequential data ordering.

Lastly, the concept of the "illusion of progress" highlights a prevalent issue in the field: the tendency to demonstrate methodological advancements primarily through the introduction of new classifiers and reliance on accuracy metrics. Some critiques this practice, suggesting a broader evaluation methodology that encompasses more than just classifier performance, where they emphasize the necessity of expanding evaluation criteria beyond accuracy alone to gain a more comprehensive understanding of model performance and methodological advancements.

In addition to that, several critical metrics have been identified to assess model performance effectively before and after drift events. These include Drift Detection Delay, or Latency, which measures the elapsed time between the occurrence of a drift and its detection by the monitoring system. Minimizing this delay is essential to enable rapid response to shift in data distribution, with the calculation explicitly defined as the difference between detection and drift occurrence times as shown in Equation 2, where is the time when the drift detection method signals the drift, and is the time when the drift actually occurs.

$$D_{delay} = t_{detection} - t_{drift} \tag{2}$$

The Magnitude of the drift aims to calculate the distance between two concepts and quantify the severity of it, like Latency, but in this case, they do not specify the measure of distance to be used, identified as D, where t and u identify the concepts in different times. The formula can be seen in Equation 3.

$$Magnitude_{t,u} = D(t, u) \tag{3}$$

Another metric is the Duration of the drift, measuring the elapsed time over which a period of drifts occurs. The formula shown in Equation 4 aims to calculate this duration based on the starting time t, and the ending time u.

$$Duration_{t,u} = u - t \tag{4}$$

A statistical measure, called the Hellinger Distance, is widely used to quantify the similarity or difference between two probability distributions, ranging between 0 and 1, where 0 indicates perfect similarity and 1 maximum dissimilarity. The formula is presented in Equation 5, where the P and Q values represent the probability distributions being compared, and the probabilities of the i^{th} event in each distribution respectively.

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{n} \left(\sqrt{p_i} - \sqrt{q_i}\right)^2} \tag{5}$$

The Path Length aims to calculate the path that the drift traverses during a period of drift. It is defined as the cumulative deviation observed during that period. The formula is bounded with the Drift Magnitude as can be seen in Equation 6, where D represents the distance between data distributions at two consecutive time steps, and t represents the time interval of these distributions.

$$PathLen = \sum_{i=1}^{n} D(P_{t_i}, P_{t_{i-1}}) \tag{6}$$

A critical measure that helps to quantify the velocity that the distribution is changing at a given time t is the Drift Rate, where the base formula can be seen in Equation 7, where given two distributions, and, at a given time t, the distance metric is defined as D and the interval between the two points is provided as.

$$Drift\ Rate = \frac{D(P_{t+\Delta t}, P_t)}{\Delta t} \tag{7}$$

This is further simplified into an Average Drift Rate by integrating the Path Length over the drift duration, where the start and end times define the integration interval, as shown in Equation 8.

$$Avg\ Drift\ Rate = \frac{PathLen}{t_2 - t_1} \tag{8}$$

On the other hand, the Local Drift Degree (LDD) helps to quantify regional density discrepancies between two different sample sets, thereby, identifying density increased, decreased, and stable regions. Unlike global drift measures, which assess overall distribution changes over larger periods, LDD specifically evaluates short-term, localized shifts, making it suitable for precise and timely drift detection. The formula commonly used can be defined as the distance between two probabilities, and often it is expressed using measures such as Kullback–Leibler divergence, Jensen–Shannon divergence, or statistical Kolmogorov–Smirnov (KS). It is globally expressed as seen in Equation 9.

$$\delta_W = \frac{\frac{|B_W|}{n_b}}{\frac{|A_W|}{n_a}} - 1 \tag{9}$$

Denote the feature space as W, where in an ideal situation $|A_W|$ and $|B_W|$ represent the number of data instances in W that belong to A, B, and represent the total number of instances in A and B respectively.

Similarly, the Margin Density (MD) assesses the density of data points near the decision boundary, also known as the margin, where the classifier is uncertain and tries to guess based on its learned information. A higher density indicates that the model is struggling to correctly classify incoming data, while the lower the density indicates a clearer separation between classes is presented, resulting in the model correctly classifying the new incoming data. The formula for a given margin can be seen in Equation 10.

$$MD(\tau) = \frac{\text{Number of data points within margin}\tau}{\text{Total number of points}} \tag{10}$$

The Conditioned Marginal Covariate Drift is the weighted sum of the distances between each of the conditional probability distributions, given as P, over the possible values of the covariate attributes for each specific class determined by y, from a period from t to u, where the weights are the average probability of the class over the two time periods. The formula can be seen in Equation 11.

$$\sigma_{t,u}^{X|Y} = \sum_{y \in Y} \left[\frac{P_t(y) + P_u(y)}{2} \cdot \frac{1}{2} \sum_{x \in X} |P_t(\tilde{x}|y) - P_u(\tilde{x}|y)| \right] \tag{11}$$

where X denotes the covariate (feature) space and Y denotes the class-label space. The variables $x \in X$ and $y \in Y$ represent individual covariate values and class labels, respectively.

In the same study, the Posterior Drift is identified where for each subset of the covariate attributes, there will be a probability distribution over the class labels for each combination of covariate values, at each period. This can be calculated as the weighted sum of the distances between these probability distributions, where the weights are the average probability over the two periods of the specific value for the covariate attribute subset as seen in Equation 12.

$$\sigma_{t,u}^{Y|X} = \sum_{x \in X} \left[\frac{P_t(x) + P_u(x)}{2} \cdot \frac{1}{2} \sum_{y \in Y} |P_t(y|x) - P_u(y|x)| \right] \tag{12}$$

The study by Anderson, Koh, and Dobbie (2018) introduces the Lift-per-Drift (LPD) aiming to measure drift detection performance through its impact on classification accuracy and penalized by drifts detected on a dataset, more specifically, it measures the improvement in predictive performance, considered "lift," that is achieved by a model after detecting and adapting to a concept drift event. It is based on a performance metric, such as accuracy or error rate. The formula can be seen in Equation 13, where n indicates the number of detected drift events, and both performances just after and before adaptation following the i^{th} drift detection event.

$$LPD = \frac{\sum_{i=1}^{n} \left(Performance_{post-drift_i} - Performance_{pre-drift_i} \right)}{n} \tag{13}$$

Finally, the Novel Precision Rate N_{Prc} measures the proportion of correctly identified novel category samples among all samples classified as novel by the model. Equation 14 presents the corresponding formula, where T_{nov} denotes the number of samples correctly detected as novel, and $T_{nov} + F_{exredistnov}$ represents the total number of samples predicted as novel, including both correct and incorrect detections.

$$N_{Prc} = \frac{T_{nov}}{T_{nov} + F_{existnov}} \tag{14}$$

When evaluating algorithmic efficiency, several mathematical benchmark functions are frequently employed. The Rastrigin Function is notable for its numerous local minima, making it ideal for assessing global search capabilities. The Ackley Function, characterized by its flat regions and central hole, poses significant convergence challenges. Additionally, the Rosenbrock Function is particularly valuable for testing optimization on a non-convex, curved surface, whereas the Schwefel Function, with its multiple local minima, provides a rigorous evaluation framework for global optimization algorithms.

Synthetic data streams have also been developed specifically to test algorithm adaptability and performance. The SEA Generator produces concepts featuring incremental drifts, ideal for classification algorithm assessment. Meanwhile, the Hyperplane Generator simulates evolving hyperplane-based concepts, which are essential for adaptive learning evaluation. Other generators include the Circles Generator, designed to create circular class boundaries useful in classification boundary detection, and the SINE Generator, which produces sinusoidal decision boundaries to rigorously test classifier adaptability. The MIXED Generator further enhances testing capabilities by combining multiple concept drifts within data streams.

Benchmarking using real-world datasets remains crucial for realistic and comprehensive algorithm evaluation. The UCI Machine Learning Repository (Kelly, Longjohn, and Nottingham 2023) provides widely used datasets such as the Wine Quality Dataset (Cortez ACP 2009) for regression and classification tasks, the Wisconsin Breast Cancer Dataset (Mangasarian and Wolberg 1993) for binary cancer diagnosis classification, the E. coli Dataset (Nakai 1996) for multiclass biological classification tasks, and the Yeast Dataset (Nakai 1991), which challenges classifiers with multiple protein localization classes. Similarly, the KEEL Repository (Alcalá-Fdez et al. 2011) offers specialized datasets beneficial for evaluating ML algorithms, particularly in evolving feature selection and algorithm benchmarking.

The WEKA Datasets repository is widely integrated into the WEKA (Witten et al. 2016) platform and facilitates convenient benchmarking across various domains, including classification, regression, and clustering tasks. In addition, large-scale and specialized datasets such as the Forest Cover Dataset (Blackard 1998), used for predictive analytics in forest classification, and the Electricity Market Dataset, essential for real-time predictive modeling of electricity price movements, enhance real-world applicability. Other significant datasets include the Human Activity Recognition (HAR) Dataset (Logacjov et al. 2021) for sensor-based activity classification, the Poker Hand Dataset (Cattral and Oppacher 2002) for evaluating algorithms via poker hand recognition, the Airlines Dataset for airline delay predictions, the Air Quality Dataset (Vito 2008) for environmental forecasting, the Rialto Bridge Dataset for anomaly detection in structural health monitoring, and the cybersecurity-focused KDDCup99 Dataset (W. Fan and Stolfo 1999).

OpenML (Vanschoren et al. 2014) is an open-source ML platform that provides almost 6.4k+ datasets (most of the datasets mentioned in this paper can be easily loaded using the OpenML Python API) in uniform standards along with standardized tasks and workflows to enable reproducibility and transparent research. It tries to future proof the availability of datasets by allowing versioning and allowing users to link

their experiments to these datasets through easily reproducible pipelines. AMLB (AutoML Benchmark) (Gijsbers et al. 2024) is a tool that builds on OpenML to facilitate easier benchmarking of models on tabular data. A special tag that links the two are benchmark datasets, a way of stating that a certain dataset is important for a subclass of tasks.

In conclusion, we should mention that in the domain of computer vision, datasets such as MNIST (Deng 2012) and CIFAR (Krizhevsky 2009) are extensively used. MNIST serves as the standard benchmark for handwritten digit classification, while CIFAR presents diverse natural image data, challenging object recognition and classification algorithms across multiple categories, making it a common benchmark dataset for any kind of image classification model.

Limitations of current evaluation metrics

The observations presented here concern the methodological constraints associated with several evaluation metrics commonly used in EML. They are included at the end of this section because these considerations refine the interpretation of the metrics introduced above, rather than reflecting the intrinsic learning challenges examined in later sections. Their placement here emphasizes that the limitations pertain to the design and validity of empirical evaluation protocols rather than to the conceptual taxonomy of EML itself.

Distance-based measures such as the Hellinger distance in Equation 5 rely on discretized approximations of continuous probability distributions. When applied to high-dimensional data, histogram-based estimates become sparse and unstable, which substantially reduces the reliability of the resulting distance computation. As a consequence, the Hellinger distance is most suitable in low-dimensional feature spaces or when used in conjunction with dimensionality reduction techniques that yield compact representations of the data.

Metrics that quantify post-drift recovery, such as the Lift per Drift (LPD) measure in Eq. 13, offer a distinct but equally important perspective. Because drift typically induces an immediate decline in model performance, the LPD value reflects the extent to which the model is able to recover through adaptation. It, therefore, serves as an indicator of the resilience and adaptability of the learning system rather than a direct measure of absolute predictive quality. This distinction is especially relevant in evolving environments where the difficulty or structure of the target concept may change over time, making raw accuracy an incomplete indicator of model competence.

These considerations underscore the importance of selecting evaluation metrics that are well aligned with the statistical and temporal properties of evolving data. Careful interpretation of these measures is essential for drawing valid conclusions about the behavior of adaptive models in non-stationary environments.

EML challenges

This section provides a deeper understanding of the different challenges involved in EML, that traditional static ML systems have not encountered. They can be divided into four main categories based on the extent of the challenge, where we can find Data Drift, Concept Drift, Catastrophic Forgetting, and Skewed Learning. From the extensive literature reviewed for this paper, it has been demonstrated that most current surveys focused on individual challenges, whereas only Sulaiman et al. (2025) provide a study on all the challenges associated with online learning with full feedback. Figure 1 provides a taxonomy of EML challenges.

Data drift

This challenge arises when the statistical properties of the input features change over time, as the data that it encounters has changed from the initially used during the training, impacting directly the long-term general performance of the models.

Between the surveys covered for this topic, Palli et al. (2022) investigated the performance of existing drift detection methods on multi-class imbalanced data streams with different drift types and proposed DDM as the best method based on the average F1-score. Ashok, Ezhumalai and Patwa (2023) compared and evaluated the main approaches in time-series models, providing a taxonomy and in-detail evaluation metrics, and Gemaque et al. (2020) proposed an overview of unsupervised methods.

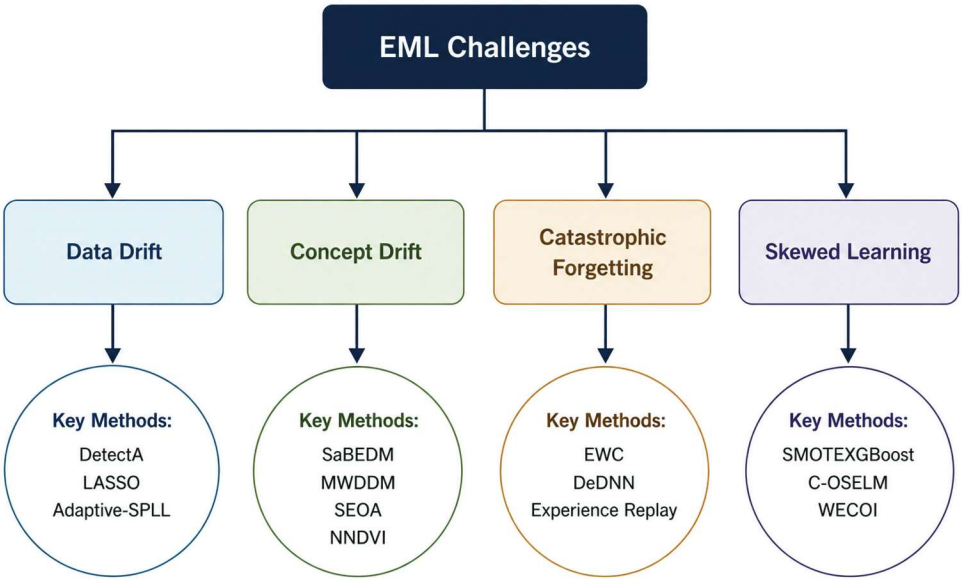

Figure 1. Taxonomy.

Unsupervised detection methods

Table 1 provides a summary of the methods examined for handling data drifts, where a study proposed the Adaptive-SPLL (Klein and Verbeke 2020) following the suggestion of using K-means clustering to estimate the density of a Gaussian Mixture model and also introduced adaptive window sizes, where the size of the windows grows until a drift is detected, or max size is reached, then it will split continuously into smaller windows to detect the actual starting time of the current drift, reusing the K-means centroids from the last window. This solution has been evaluated using an industrial dataset obtained from the NASA, containing C-MAPSS for degradation, showing an increase in accuracy and reduced computation time, but similar detection delays to non-adaptive solutions.

The DetectA (Escovedo et al. 2018) proposed is a proactive drift detection approach that identifies abrupt drifts in the data stream before they occur, using a Multivariate Hypothesis test to compare the statistics that could indicate the presence of drift. Two different experiments were performed, where the first aimed to identify the sensitivity of the approach by computing its false positives and false negatives, where their results suggest that this detector is more efficient for high-dimensional datasets with intermediate-sized blocks, using datasets with any proportion of drifts and imbalanced binary classes. The second one aimed to measure the performance of the

Table 1. Drift detection methods.

Ref	Category	Method	Drift	Datasets
Klein and Verbeke (2020)	Unsupervised	Adaptive SPLL	N/A	Synthetic in-house dataset for lab testing and NASA data with C-MAPSS for degradation
Escovedo et al. (2018)	Unsupervised	DetectA	Abrupt	SEA Concepts, Nebraska, Electricity, Cover Type and Poker hand
Suprem, Musaev, and Pu (2019)	Unsupervised	LITMUS-adaptive	N/A	Not specified
Porwik et al. (2025)	Supervised	Least Absolute Shrinkage and Selection Operator (LASSO)	All	Synthetic (MOA generated), Phishing, Electricity, Ozone, Arrhythmia, Poker hand, Rialto Bridge timelapse, Outdoor
Senarathna et al. (2023)	Supervised	N/A	Simultaneous	MNIST, CIFAR10
Ye, Kim, and Jeon (2025)	Supervised	N/A	Simultaneous	Social-sensor data (2014–2018; Facebook, Twitter), Rainfall from NOAA, Earthquake data from USGS, Landslide predictions from NOAA and USGS
Krawczyk, Pfahringer, and Woźniak (2018)	Semi-supervised	RAND++, VAR-UN++, R-VAR-UN++	All	Airlines, Electricity, Forest Cover, RBF, Hyperplanes, Tree

detector using a multi-layer perceptron (MLP) as the baseline model along with different datasets, such as SEA concepts, Nebraska, Electricity, Cover Type, and Poker Hand. Different versions of the method are compared, such as no detection, reactive detection, and proactive detection, where the latter approaches performed better and help to improve the general performance of the model, the no detection versions were faster in computational time as expected.

Other approaches take an adaptive data augmentation approach, such as the LITMUS-adaptive (Suprem, Musaev, and Pu 2019) that extends the LITMUS-static by applying a detector monitoring when a signal exceeds a given threshold, applied for physical event detection using data for social networks, then a new classifier is trained using the previous data window available, and previous models get their parameters updated. In this method, all the previously created models are stored in a database as key-values of the new and previous model, along with the centroids as the key for a classifier. The prediction is then obtained by producing voting between the ensemble of models. The experimental results showed that the adaptive version outperforms the static one with almost 350% more landslide events identified, where the F1-score reached a 98% against the 76.2 of the static.

Supervised detection methods

The LASSO (Porwik et al. 2025) proposed a feature rank drift detection procedure based on analyzing changes in feature ranks across adjacent chunks of data, and monitoring its fluctuations by comparing them to the reference feature ranks. When a change is identified, an alarm gets triggered and the classifier is retrained on the new data chunk, updating the baseline ranks to reflect the new distribution. Different tests were conducted, the ones using Synthetic showed that the proposed solution outperformed DDM, EDDM, ADWIN, and PCA-FDD for the Abrupt and Recurring data drift types, while the result on the real-world datasets seems to deliver almost as the other compared methods, but lowering considerably the computational time in almost every scenario.

A method that measures the magnitude of image classification tasks on noisy weather effects tasks (Senarathna et al. 2023) that uses MNIST and CIFAR10 images, and stores the prediction probabilities of each class in the network, helping to identify all kinds of drifts and estimate the magnitude with high accuracy, but has not been compared with other methods. This approach has been extended by Ye, Kim, and Jeon (2025) using cumulative distribution functions (CDFs) of output probabilities instead of base prediction probabilities. In the experiments, the proposed method used CIFAR10 along with a RandomRain function to demonstrate that using the two drift magnitudes as a combined pair reduces the overestimation of drift levels, helping to increase the accuracy.

Semi-supervised detection methods

Similarly, Krawczyk, Pfahringer, and Woźniak (2018) introduced strategies that use a drift detector module, such as ADWIN2, that uses the labeled samples as the active learning, and when the accuracy of the classifier starts decreasing, another classifier is trained on the background using arriving objects, so in case of the change is finally detected, the classifier is replaced. The Random strategy++ randomly draws instance labels with probability equal to the assumed budget, this is done by improving the label query if a change is detected, resulting in increasing the labeling probability according to the output of the drift detector. The Variable uncertainty strategy++ is based on monitoring the certainty of classifiers, aiming to label the least certain instances within a time interval, and resulting in a balanced budget over the time. In this strategy, the threshold is rapidly reduced once the alarm or the drift is detected to allow the gathering of the highest number of labeled objects to quickly adapt a new model. Finally, the Randomized version of the previous strategy, where the threshold is simply randomly selected. Experiments between the different versions were conducted and compared with a fully supervised classifier using different benchmark datasets, such as Airlines, Electricity, Forest Cover, RBF, and Hyperplanes, and evaluated using prequential accuracy to determine if these strategies are closer in accuracy to the fully labeled classifier, where the results showed that the only in the electricity datasets the proposed active learning strategies were similar, even with reduced budget, while the remaining datasets showed a significant increase in accuracy when the feedback received from the drift detector is utilized by the label query.

Key insights

The methods examined in this section highlight a shift from purely reactive drift detection toward more anticipatory and robust strategies. Approaches such as **DetectA** exemplify proactive monitoring by examining changes in the input distribution, while **DDM** remains representative of traditional error-driven detection. Techniques based on feature importance, such as **LASSO**, provide a complementary perspective by identifying structural changes in $P(Y|X)$ that are less sensitive to noise. Methods including **Adaptive SPLL** further illustrate the value of flexible windowing in industrial contexts in which drift often unfolds gradually. Together, these methods suggest that effective data drift detection requires coordinated sensitivity to both input variation and concept level change, supported by mechanisms that remain stable under noise and gradual evolution.

Although data drift constitutes a fundamental challenge in EML, the literature identified in this review is notably smaller than that for concept drift and catastrophic forgetting. This disparity is reflected in both the number of surveys and primary studies included in our analysis. One possible explanation is that data drift is frequently treated as a precursor to concept drift or is addressed indirectly through drift-detection frameworks rather than as a standalone research problem. Consequently, the limited number of dedicated data-drift studies should not be interpreted as a lack of practical relevance, but rather as evidence of an underexplored area that warrants further investigation.

Concept drift

This type of challenge represents the changes in the underlying relationship between input and target variables, changing the conditional distribution of target variables given the input data.

Types of drift

Before exploring drift in detail, the types of drift are outlined below. Figure 2 shows the different types of concept drift. Sudden Drift: It is the shift or change in the data distribution that has occurred abruptly and unexpectedly. In general, it can be disruptive and challenging to detect. For example, a fraud detection model fails after an unexpected policy change causes a spike in new fraud tactics overnight.

Gradual Drift: This kind of shift or change happens over an extended period, and its effects cannot be immediately noticeable. For example, fraudsters slowly adapt their techniques over months, subtly reducing the model's effectiveness without immediate signs.

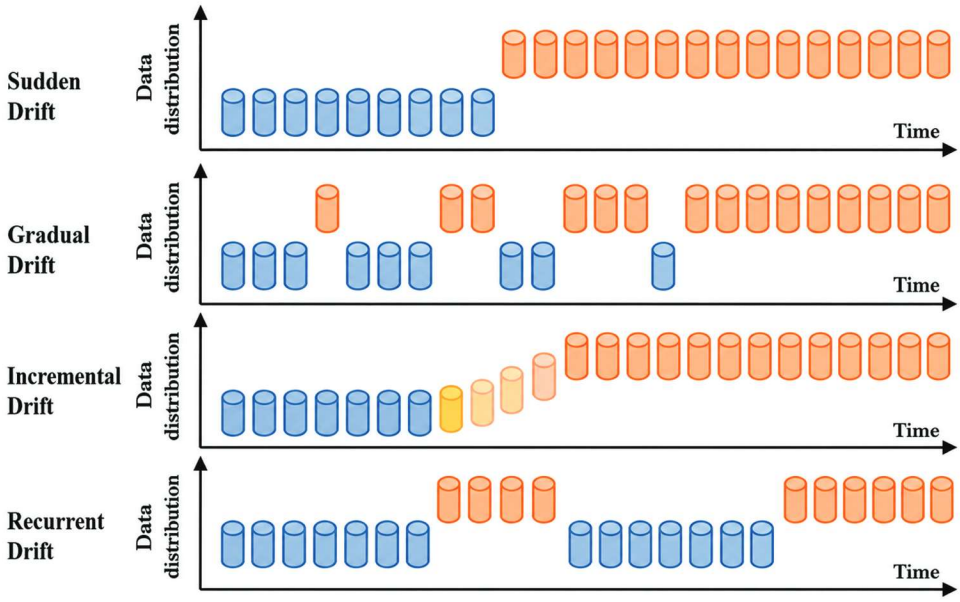

Figure 2. Types of concept drift.

Incremental Drift: It is the change in the relationship between the input and the output that happens slowly and gradually. For example, the types of fraudulent behavior slowly evolve with minor adjustments, gradually changing the relationship between transaction features and fraud labels.

Recurring Drift: This type of change tends to appear and disappear cyclically, in a patterned way, resulting in periodical changes that are not constant. For example, fraud patterns spike during major shopping seasons like Black Friday each year, following a predictable cycle.

Comprehensive survey reviews and taxonomies

A wide range of survey reviews have been covered during the production of this study, where it is necessary to mention the work effectuated by Barros and Santos (2018) and de Barros Rsm et al. (2019) with more than 50 method covered and produce a large-scale comparison, Iwashita and Papa (2019) with 71 reviews covering methods and public available datasets, Lu et al. (2019) introduced an extensive survey review with over 130 methods covering all areas of research, 10 synthetic datasets, and 14 public available dataset (Elkhawaga et al. 2020) also produced an extensive literature review in concept drift analysis and proposed the framework called Concept Drift Analysis in Process Mining (CONDA-PM) that aims to guide users in evaluating the maturity of an approach. Moleda, Momot, and Mrozek (2020) analyzed methods, techniques, and approaches applied in coal-fired power plants for fault prediction.

Bayram, Ahmed, and Kassler (2022) proposed a taxonomy classification for all drift detectors that is quite complete and well-presented. Hinder, Vaquet, and Hammer (2022) presented a comprehensive review of metric choices for concept drift detection, comparing the existing ones and proposing new choices demonstrating their suitability. Hinder, Vaquet and Hammer (2023) provide a detailed survey of unsupervised methods, providing a taxonomy of approaches and detailed mathematical definitions. Sakurai et al. (2023) survey is oriented to guide in the selection of the best concept drift detector by testing and benchmarking them in their work. Suárez-Cetrulo, Quintana, and Cervantes (2023) provide an extensive survey of supervised and unsupervised detectors that covers passive and active detectors, but also adapters. Xiang et al. (2023) summarized concept drift adaptation methods for deep learning frameworks. Hovakimyan and Bravo (2024) proposed a literature review that adheres to PRISMA guidelines aiming to provide a valuable resource for future research.

Werner et al. (2024) focus on the performance of unsupervised concept drift detectors, their computational complexity, and strategies for benchmarking them. Costa et al. (2025) focused on the impact of four descriptors (severity, recurrence, frequency, and speed) and conducted experiments on five benchmark datasets with 32 descriptor variations, resulting in a total of 1440 combinations. Lukats et al. (2025) explore the availability of fully unsupervised concept drift detectors by analyzing the performance of ten of them in real-world data streams.

Sliding window and statistical comparison methods

Table 2 summarizes the widely used type of methods to mitigate this issue, where the commonly approach is single or multiple sliding windows are applied to identify subsets of data from continuous data streams, moving forward as the new data arrives, helping the model to compare the statistics received in two or more different data moments and adapt quickly to new conditions. Among these sliding window methodologies, Monte Carlo Dropout in Baier et al. (2022) under the Uncertainty Drift Detector (UDD) method without relying on true labels has showed that this application of ADWIN can increase considerably the accuracy, in comparison with KSADWIN and its base version, when evaluated using a multiple benchmark datasets, and metrics such as Mean Time Detection (MTD), Missed Detection Count (MDC), Root Mean Squared Error (RMSE), and Matthews Correlation Coefficient (MCC).

The double-window mechanism to handle concept drifting data streams that calculates the standard deviation and the error rate on the current and previous windows, where other calculations have been applied between these windows, such as the neural network combined approach (Shang, Zhang, and Lu 2025) capable of distinguishing between known labeled concept drift and novelty, employing PCA and radial distance, has been evaluated using synthetic and CIFAR10 datasets, compared with CM, MMD, KDQ, and LDD by applying metrics such as number of drifts and novelty detected, and Maximum Mean Discrepancy (MMD), Local Drift Degree (LDD), Accuracy, running time.

Table 2. Summary of concept drift approaches.

Ref	Section	Category	Method	Drift	Datasets
de Lima Cabral Dr and de Barros Rsm (2018)	Sliding Windows	Supervised	FPDD, FSDD, FTDD	Abrupt, gradual	24 artificial datasets with LED, Mixed, RandomRBF, and Sine generators; Airlines, Covertype, Poker hand
Q. Li et al. (2021)	Sliding Windows	Semi-supervised	Incremental Semi-supervised Extreme Learning Machine (MIS-ELM)	N/A	Adult, Census, Credit, NSL-KDD2, NSL-KDD5, Mushroom, Nursery
Tanha et al. (2022)	Sliding Windows	Semi-supervised	Conformal prediction for semi-supervised classification (CPSDDS)	Various	SEA, STAGGER, AGRAWAL, RBF, Waveform, Elec2, Spam, Vehicle
Bu, Alippi, and Zhao (2018)	Sliding Windows	Unsupervised	Least Squares Density-Difference (LSDD)	Abrupt	N/A
de Mello Rf et al. (2019)	Sliding Windows	Unsupervised	Plover	N/A	Synthetic, S&P500, Electricity, Airlines
Gözüac¸ık et al. (2019)	Sliding Windows	Unsupervised	D3	N/A	Electricity, Covertype, Poker hand, Rialto, Rotating Hyperplane, Moving RBF, Interchanging RBF
Shang, Zhang, and Lu(2020)	Sliding Windows	Unsupervised	Unsupervised Concept Drift Detection (UCDD)	Various	Synthetic
Souza, Chowdhury, and Mueen (2020)	Sliding Windows	Unsupervised	Image-based Drift Detector (IBDD)	Various	Heartbeats, Flying insects, Sensor Posture, StarLightCurves, UWaveGestureLibrary, Yoga
Xuan, Lu, and Zhang (2020)	Sliding Windows	Unsupervised	Bayesian Nonparametric Drift Detector (BNDDM)	Sudden drifts	Keystroke, Insects, Arabic, Bike, Posture
Ömer Gözüaçık and Can (2021)	Sliding Windows	Unsupervised	One-Class Drift Detector (OCDD)	N/A	13 Synthetic and real-world datasets
Wan and Wang (2021)	Sliding Windows	Unsupervised	Clustered Statistical Test for Drift Detection Method (CSDDM)	N/A	SEA, Hyperplanes, Elec2, Covertype
Baier et al. (2022)	Sliding Windows	Unsupervised	Uncertainty Drift Detection (UDD)	N/A	Friedman, Mixed, Air Quality, Bike Sharing, IncAbr, IncReo, KDDCup99, Gas sensor, Elec2, Rialto
Balestra, Li, and Müller (2023)	Sliding Windows	Unsupervised	SlidSHAPs	N/A	LED, ADD, MUL, COE, AND, OR, XOR, MIX, BC, PH, KDD, MSL
Q. Fan et al. (2023)	Sliding Windows	Unsupervised	Concept Drift Detection based on Jensen–Shannon Divergence	N/A	Artificial, Real-world
Zheng (2019)	Sliding Windows	N/A	Labelless Concept Drift Detection and Explanation (L-CODE)	N/A	SEA, Electricity, Phishing, Adult, Rabobank
Baidari and Honnikoll (2021)	Sliding Windows	N/A	Bhattacharyya Distance-based Concept Drift Detection Method (BDDM)	Gradual, abrupt	Agrawal, LED, RBF, Electricity, Shuttle, Connect-4, Diabetes, Airlines, Poker hand, Covertype, Pen digits, Gesture, Codrna
Cavenaghi et al. (2021)	Sliding Windows	N/A	f-Discounted-Sliding-Window Thompson Sampling (f-dsw TS)	All	Baltimore Crime, Insects (variants), Local news, Air Microbes
Naqvi, Rehman, and Islam (2022)	Sliding Windows	N/A	WinDrift (WD)	N/A	4 real and 10 reproducible synthetic datasets
Chen, Han, et al. (2023)	Sliding Windows	N/A	Multi-level Weighted Drift Detection Method (MWDDM)	Abrupt, gradual	SINE, MIXED, CIRCLES, LED, Elec2, Forest Covertype, Poker hand
Agrawal et al. 2020)	Ensemble-based	N/A	LightGBM with drift detection	N/A	Power grid
AlQabbany and Azmi (2021)	Ensemble-based	N/A	Enhanced ARF	Various	LED, SEA, Agrawal, RTG, RBF, Hyperplane, Amazon customer review, Hotel Arabic-Reviews, COVID-19
Angbera and Chan (2024)	Sliding Windows	N/A	sliding adaptive beta distribution model (SABeDM)	N/A	N/A
Cassales et al. (2021)	Ensemble-based	N/A	Mini-batching strategy for bagging ensembles	N/A	Airlines, GMSC, Elec2, and Covertype
Castellani, Schmitt, and Hammer (2021)	Sliding Windows	N/A	Task-Sensitive Concept Drift Detector with Constraint Embedding	N/A	RBF and MovingRBF, Adult, bank, digits08, digits17, musk, phising, wine
Celik and Vanschoren (2021)	Other	N/A	Adaptation Strategies	Gradual, sudden, mixed	Airlines, Elec2, IMDB, Vehicle, SEA, Hyperplane
Chambers, Gaber, and Ghomeshi (2025)	Ensemble-based	Supervised	DeepStreamEnsemble (DSE-DD)	Abrupt, gradual, recurrent	Fashion-MNIST, CIFAR10, CIFAR100

(*Continued*)

Table 2. (Continued).

Ref	Section	Category	Method	Drift	Datasets
Chapelin et al. (2025)	Ensemble-based	Supervised	Data-driven Drift Detection and Diagnosis Framework	Not specified	Power private data
Dosovitskiy et al. 2021)	Meta-learning	N/A	ViT	Not specified	Not specified
Folino, Guarascio, and Papuzzo (2019)	Ensemble-based	Supervised	Fractal	Abrupt, gradual	Sine, Hyperplane, Line, Plane, Circle
Friedrich, Sawabe, and Hein (2023)	Other	Unsupervised	Variational Autoencoders (VAE)	Sudden, gradual, incremental, recurring	Synthetic dataset; sensor data from 20 participant apartments
Guo et al. (2022)	Sliding Windows	Supervised	Concept Drift Type Multi-sliding Windows (CDT_MSW)	Various	Banana, Titanic, Thyroid, Diabetes, Breast_cancer, Heart, German, KDDCup99, MITface, USPS, RBFblips, Tree, LED, SEA
Guo et al. (2025)	Ensemble-based	Supervised	Accelerating Convergence Method of Concept Drift Based on Knowledge Transfer (ACC_KT)	Sudden, gradual	Hyperplane, SEA, LED, RBFblips, Tree, Electricity, Weather, KDDCup99, Covertype
Guo et al. (2023)	Other	Supervised	Noise Reduction Soft Boundary (NRSB)	Sudden, gradual	Hayes-Roth, Page Blocks, Yeast, Poker Hand, Forest, KDDCup99
Guo, Zhang, and Wang (2021)	Ensemble-based	Supervised	Selective Ensemble-based Online Adaptive DNN (SEOA)	Various	SEA, Hyperplane, RBF, LED, Tree, Elec2, KDDCup99, Covertype, Weather
Han, Meng, and Li (2024)	Sliding Windows	N/A	Variance Feedback Drift Detection Method (VFDDM)	Abrupt, gradual	Sine, Mixed, Circles, LED, Elec2, Forest Covertype, Poker Hand, Spam
Iwashita et al. (2019)	Ensemble-based	Supervised	Optimum-Path Forest (OPF)	Abrupt, gradual	Hyperplane, Usenet, SEA, Covertype, Elec2, Poker Hand
Jaworski, Rutkowski, and Angelov (2020)	Other	Unsupervised	Concept Drift Detection Using Autoencoders	Gradual, sudden	Synthetic RBM
Kaminskyi, Li, and Müller (2022)	Other	Unsupervised	Autoencoder-based Concept Drift Detector (AECDD)	Sudden, incremental	Sine (incremental and sudden)
Kim and Jeon (2025)	Meta-learning	N/A	Use of Pre-trained ResNet and Vision Transformer Models	Not specified	ImageNet-1k
Komorniczak, Zyblewski, and Ksieniewicz (2022)	Ensemble-based	Supervised	Statistical Drift Detection Ensemble (SDDE)	Gradual, sudden, incremental	Elec2, Banknote, Diabetes, Wisconsin
Korycki and Krawczyk (2019)	Ensemble-based	Unsupervised	Ensemble Drift Detection in Feature Subspaces (EDFS)	All	Hyperplane, LED, RBF, SEA, Tree
J. Li et al. (2024)	Sliding Windows	N/A	Type-Driven Lazy Drift Adaptor (Type-LDA)	All	Toy dataset generated with MOA, Elec2, Weather, Spam, Airlines, Poker Hand
Liu et al. (2018)	Ensemble-based	Supervised	Nearest Neighbor-based Density Variation Identification (NN-DVI)	Sudden, gradual, regional	Elec2, Weather, Spam
Shang, Zhang, and Lu (2025)	Sliding Windows	N/A	Novelty-aware Concept Drift Detection for Neural Networks	All and novelty	Synthetic datasets, CIFAR10
Siahroudi, Moodi, and Beigy (2018)	Ensemble-based	Supervised	MUKERS	Sudden	SynCN, KDDCup99, CoverType

The least-squares density-difference estimation approach with its two extensions for sliding window and ensemble approach types (Bu, Alippi, and Zhao 2018) that is compared with KNN, H-ICI CDT, and CPM tests in unspecified datasets by applying False-Positive rate, False-negative rate, Delay, and CT execution time, to determine that this solution works in terms of promptness and accuracy.

The Fisher's Exact test approach (de Lima Cabral Dr and de Barros Rsm 2018), based on STEPD, introduces three different methods referred as FPDD, FSDD, and FTDD, that were compared using 24 synthetic dataset with DDM, ECDD, SEED, FHDDM, and STEPD using accuracy, precision, recall, and F1-score, to determine that these methods deliver higher predictive accuracy in most datasets, or in the top three with other competitors.

Distance and divergence-based approaches

The plover algorithm utilizing Statistical Moments and the Power Spectrum to compare the difference between the data windows (de Mello Rf et al. 2019), that has been evaluated on different synthetic, S&P500, Electricity, and Airlines datasets using Standard deviation and Power Spectrum as measures, and compared with CUSUM, PHT, EWMA, SONDE, GWR, and ADWIN, showing that this method can be compared in performance with the ADWIN method.

The D3 method employing the KL divergence to measure the change between two windows (Gözüac¸ık et al. 2019) has been evaluated on Electricity, Covertype, Poker hand, Rialto, Rotating Hyperplane, Moving RBF, and Interchanging RBF using accuracy as the chosen metric and comparing the results with ADWIN, DDM, and EDDM, determining that it outperforms the baselines, yielding models with higher performances on both real-world and synthetic datasets.

The label-less drift detection (L-CODE) (Zheng 2019) analyzing changes in the space of Shapley values and the feature distribution, has been evaluated on SEA, Electricity, Phishing, Adult, and Rabobank datasets using Accuracy, Feature distribution, and Shapley distribution, against HDDDM, MD3, ADWIN, DDM, and PH, along with NB and HT as base classifiers, proving that the method works but does not outperforms other methods that require labeling.

While another method applying the Bhattacharyya distance metric (BDDM) (Baidari and Honnikoll 2021) measuring the similarities between probability distributions over time, makes use of Agrawal, LED, RBF, Electricity, Shuttle, Connect-4, Diabetes, Airlines, Poker hand, Covertype, Pen digits, Gesture and Codrna, and the metrics Bhattacharyya distance, Accuracy, precision, recall, FN, FP during evaluation, where the results obtained has been compared to DDM, ECDD, SEED, STEPD, FHDDM, WSTD, applying the Hoeffding Tree or NB as base classifiers, proving that accuracies were comparatively better than those of the other method, with slightly better results in abrupt scenarios than the gradual ones.

The combination of applying Exponentially Weighted Moving Average (EWMA) and the Jensen-Shannon Divergence (Q. Fan et al. 2023), referred as CDDDE, proposes the calculation of drifts dynamically without the use of labels. This approach has been evaluated on multitude of synthetic and real-world datasets using Accuracy as their sole metric, determining that this approach efficiently detect the concept drift, and the retrained classifier effectively improves the classification accuracy.

Adaptive windowing and weighted models

The Sliding Adaptive Beta Distribution Model (SABeDM) (Angbera and Chan 2024) combines sliding windows of different sizes with the beta distribution technique to track changes in underlying distributions of the data. It can be applied for online learning, data stream mining, and real-time monitoring systems, where it has been compared to SRP, ADWIN, DDM, and EDDM using metrics such as accuracy, precision, recall, and F1-score, and determined that it offers an effective and fast way to identify concept drifts.

The Multi-level Weighted Drift Detection Method (MWDDM) (Chen, Han, et al. 2023) introduced three levels of detection based on the predictions received, the "stable level" occurs when no drift is presented and uses a linear weighting method that aims to assign more weight to new classes and balancing them with the previous ones while updating the model. The "warning level" seeks to increase the difference in weight values between the instances in the long and short windows, and update them, to reduce the drift to stop it happening. Then the "drift level" is determined by the implementation of Hoeffding inequality or McDiarmid inequality tests, depending on the variant of the method, that is aimed to calculate the difference between both windows' average of correct predictions, highlighting the drift and resetting the classifier for the new data distribution. This approach has been evaluated on SINE, MIXED, CIRCLES, LED, Elec2, Forest Covertype, and Poker hand using Accuracy, Detection delay, TP ratio, FP ratio, and FN ratio, where the results were compared with DDM, EDDM, RDDM, FHDDM, FHDDMS, MDDM, and HDDM, determining that this method detects drift faster, maintaining a low FP and FN ratio, keeping a high accuracy with lower computational cost.

In addition, multiple small-scale windows incorporating fast variance estimation strategies and variance-based dynamic weighting (Han, Meng, and Li 2024) to adjust the weighting of the window data instances, amplifying the window mean and enhancing adaptability, in combination with the variance feedback strategy, which allows providing different mean calculation and statistical testing methods based on different variances, helping to provide a tailored to the data solution. This method has been evaluated on

Sine, Mixed, Circles, Led, Elec2, Forest Covertype, Poker hand, and spam datasets using metrics such as TP, FP, FN, detection delay (DD), and average accuracy, comparing the results with FHDDM, MWDDM_H, MDDM, WMDDM, BDDM, RDDM, DDM, HDDM_A, and HDDM_W, proving that it can effectively contend with different types of concept drift and has good robustness and generalization.

Label-free and unsupervised drift detection techniques

Other methods demonstrated the capability of current techniques for the detection of the changes, such as the case for, the UCDD method (Shang, Zhang, and Lu 2020) implying a clustering technique to label the data, and identify points located on the boundaries of the labeled data. The evaluation has been performed on synthetic datasets using drift magnitude as the sole metric, where the results are compared with CM and KDQ, determining that it outperforms the other compared methods.

The model-independent method for image classification (Souza, Chowdhury, and Mueen 2020) that converts the data contained on the windows into a 2-D representation and compares the difference between the pixel intensity of them, it has been evaluated on Heartbeats, Flying insects, Sensor Posture, StarLightCurves, UWaveGestureLibrary, and Yoga datasets using metrics such as mean-squared deviation, cost over features, number of drifts detected, and accuracy, demonstrated to be faster and more accurate than IKS, WRS, an static classifier for baseline, and dynamic classifier with persistent detector for top line.

The Bayesian Nonparametric unsupervised framework, referred by BNDM (Xuan, Lu, and Zhang 2020) utilizes Polya tree hypothesis test decomposing the data distributions into a multi-resolution representation. It has been evaluated on Keystroke, Insects, Arabic, Bike, and Posture datasets using accuracy, and detection rate as their chosen metrics to compare with KS, RDa, RDm, MD3, PH, UE, HHT-u, baseline, and topline classifiers, determining that this method is generally more sensitive to sudden and non-fixed space drifts comparing with other drift types.

A single windows method based on Multi-Armed Bandit, referred as f-dsw TS (Cavenaghi et al. 2021), enhancing the traditional Thompson Sampling approach along with the Discount factor, applying decay to past rewards, and giving more weight to recent observations. This method has been evaluated on The Baltimore Crime, Insects (Different variants), Local news, and Air Microbes using Regret, Cumulative reward, and Accuracy as their choice of metrics, and compared with other MAB methods, such as Thompson Sampling, Discounted TS, Sliding TS, min-dsw TS, mean-dsw TS, and max-dsw TS, where the proposed method emerged as the best performing MAB algorithm.

The one-class drift detector for unsupervised learning (OCDD) (Gözüaçık and Can 2021) utilizes one window to monitor deviations that may indicate concept drift. This approach has been evaluated on a sum of 13 synthetic and real-world datasets and compared with other 17 methods using accuracy and determining that it outperforms the other methods by producing models with better predictive performance.

The MIS-ELM (Q. Li et al. 2021) extends the ELM approach by applying a soft one-hot encoding on the categorical data, and operates incrementally by updating the model with incoming data to adapt to evolving patterns, and during stable periods, it leverages unlabeled data to fine-tune the existing classifier. This method has been evaluated on Adult, census, credit, NSL-KDD2, NSL-KDD5, mushroom, and Nursery datasets using accuracy and time overhead, then the results were compared with other ELM-based methods, such as RELM, SS-ELM, US-ELM, ILR-ELM, OS-ELM, OR-ELM, WOS-ELM, and ESOS-ELM, where the results proved it improves the performance of ELM methods, while the computational cost showed a trade-off.

Another unsupervised method uses PCA, clustering using K-means, and Anderson–Darling statistical test to compare and detect significant differences (Wan and Wang 2021), referred to as CSDDM. It has been evaluated on SEA, Hyperplanes, Elec2, and COVTYPE using metrics such as Accuracy, Number of drifts, and percentage of label used, TPR, FPR, ROC, and Execution time, where the results have been compared against an Static classifier, sliding window detector, and SCARGC, showing that CSDDM cannot detect the real concept drift as accurate as a supervised one, but with lower false alarm rate and competitive classification accuracy than other unsupervised approaches.

The CPSSDS method (Tanha et al. 2022) applies an incremental classifier with self-training to enhance learning from scarce labeled data by applying Conformal Prediction to identify unlabeled data points. It has been evaluated on SEA, STAGGER, AGRAWAL RBF, Waveform, Elect2, Spam, and Vehicle datasets using accuracy as the choice of metric, and compared with Amanda, Fast Compose, STDS, Sco-Forest, SCBELS

integrating HT and NB as base classifiers, where the CPSSDS showed it superiority against the other methods compared in the experiments.

The Type-LDA (J. Li et al. 2024) aims to identify the type of drift and the drift location point using a pre-trained drift type classifier on synthetic data, helping the base model to effectively select appropriate retraining strategies based on a sharing loss and maximizing the Gaussian likelihood function. It has been evaluated on Toy dataset generated with MOA, Elec2, Weather, Spam, Airlines, and Poker hand using metric such as F1-score, accuracy, running average time, and drift detection number. When compared with DWIN, DDM, EDDM, HDDM-A, HDDM-W, Page-Hinkley, KSWIN, AE, BLAST, and KUE, this approach proved that accurately identifying drift type and improve its adaptation accuracy better than the mentioned ones.

Another methods focused on time-series data employing Kolmogorov–Smirnov statistical distance, referred as WD (Naqvi, Rehman, and Islam 2022) to compare corresponding periods. It has been evaluated on 4 real datasets and 10 reproducible synthetic datasets using KS distance and accuracy as the metric chosen, and compared against five detectors, determining that it detects drift efficiently with minimal false alarms and has efficient computational resource usage. Similarly, the slidSHAPs method for multivariate time series (Balestra, Li, and Müller 2023) that employs Shapley values to quantify the contribution of each variable to the overall correlation structure of the sliding windows. It has been evaluated on LED, ADD, MUL, COE, AND, OR, XOR, MIX, BC, PH, KDD, MSL datasets using Precision, recall, F1, and Average delay, and when compared to HDDDM, and ADWIN, it proved a dominating performance on almost all datasets.

Ensemble-based drift adaptation methods

In the other side, the methods studied for the ensemble-based type, we are able to find supervised approaches such as the NN-DVI (Liu et al. 2018) that directly addresses the root causes of concept drift, consisting of three parts, where the first part follows the data modeling approach that retrieves the critical information from the dataset using a nearest neighbor-based partition schema, the second part is a distance function that accumulates regional density changes to quantify the overall discrepancy between the datasets, and the last component is a tailored statistical significance test for the distance. This approach has been evaluated on Elec2, Weather, and Spam datasets using accuracy, detected drifts, False detection rate, and False alarm rate, where it was compared with MMD, KL, CM, SAM-KNN, HDDM, ADW-Vol, AUE2, HAT, and ADWIN, showing that NN-DVI is capable of detect drifts accurately with similar performance.

Another method based on multi-kernels with boundaries and centers of classes in the feature space (Siahroudi, Moodi, and Beigy 2018), called MUKERS, where the data gets mapped into a Hilbert space in a search of linear relationships, and the multiple kernels actuate as different classifiers in an ensemble. This approach has been evaluated on SynCN, KDDCup99, and CoverType datasets using Accuracy, running time and memory as their metrics of choice, and compared with Loce, CLAM, ECSMiner, and SvsClass, where its performance was similar in the accuracy level, with a very high computational complexity, causing delays. A combination of DDM or EDDM with LightGBM (Agrawal et al. 2020), proposed for cybersecurity purposes to detect and adapt to concept drift, where the data received is passed into the base learner, and its output is passed to the detector, which will classify the input as a warning, no drift level or drift level as soon as the data arrives. If no drift level, the algorithm continues to train from the data, if the warning level is raised, the data instances are stored, so when the drift level is recognized, a new model is trained using the same data instances that were stored during the warning level. This approach has been evaluated on Power grid data using metrics such as Precision, Recall, F1, and Accuracy, then compared to Adaboost, XGBoost, SAE DL, NNGE+STEM, and a CNN, where the results showed an increase on the accuracy, that reached 97.73%. An improvement of the Adaptive Random Forest algorithm that implements the Resampling Effectiveness Measure (AlQabbany and Azmi 2021) provides an important contribution to the improvement of these algorithms combining accuracy and execution time to evaluate the efficiency of the methods. In addition, they also proposed an empirical solution to select the parameter for the Poison distribution used during the resampling process. This approach has been evaluated on LED, SEA, Agrawal, RTG, RBF, Hyperplane, Amazon customer review, Hotel Arabic-Reviews, and COVID-19 datasets using metrics such as Resampling effectiveness based on accuracy and execution time. It has been compared with ARF to determine that it exhibited considerable improvement than the base model.

A mini-batching strategy (Cassales et al. 2021) aimed to increase the performance of bagging ensemble algorithms by aggregating incoming data in batches before it gets processed, improving memory access locality and reducing cache misses, helping to speed up to 5 times faster. This approach has been evaluated on Airlines, GMSC, Elec2, and Covertype datasets using metrics such as distance analysis, Memory footprint, cache usage, accuracy, precision, and recall, where the results are compared with Lbag, ARF, SRP, OBAdwin, OBASHT, and OB proving it can improve the energy efficiency of bagging ensembles, and alleviate the memory bottleneck, increasing the speedup to some extent.

The SEOA (Guo, Zhang, and Wang 2021) framework enhances model convergence by integrating shallow and deep features through adaptive depth units, which dynamically manage information flow based on changes in adjacent data streams. Each adaptive depth unit functions as a base classifier within an ensemble, with weights adjusted in real-time according to individual performance losses. Furthermore, SEOA employs a dynamic selection mechanism that responds to data fluctuations, balancing stability, and adaptability. It has been evaluated on SEA, Hyperplane, RBF, LED, Tree, Elec2, KDDCup99, Covertype, and Weather using average accuracy, accuracy, final cumulative accuracy, and recovery speed under accuracy (RSA), then compared with DNN-2, DNN-4, DNN-8, DNN-16, Resnet, Highway, and HBP, where it has proven that it can effectively contend with different types of concept drift and has good robustness and generalization. The SDDE method (Komorniczak, Zyblewski, and Ksieniewicz 2022) expands the previously proposed SDDM method, which employs an ensemble of statistical tests, and different measures that are calculated based on the Kernel Density Estimation to monitor different aspects of the data distribution that can lead to drifts. It has been evaluated on Elec2, Banknote, diabetes, and Wisconsin using Drift magnitude, conditioned marginal covariate drift, and accuracy, then compared with DDM, EDDM, ADWIN, and HDDM, where the results obtained clearly outperformed the compared methods. The ACC_KT approach (Guo et al. 2025) extracts valuable knowledge from pre-drift data, transferring it to post-drift data by keeping historical model data that contains the key data to be passed to the ensemble classifiers, so every time a new one is trained, the new data is passed along with the key data. When drift occurs, a new model is trained, in case the total number of classifiers is lower than the maximum, a new model is then added, if the maximum is reached, the new model replaces the classifier that showed the lowest performance. It has been evaluated on Hyperplane, Sea, LED, RBFblips, Tree, Electricity, Weather, KDDcup99, and Covertype using average accuracy, average AIF, and average recoverability, where the results compared with DNN-4, ResNet, Highway, HBP, DenseNet base classifiers with the methods DWM, SRP, DWCDS, ARF, SAL, and DTEL showed that this approach provides an acceleration effect, but still require more efficiency on different types of drifts and optimization.

It is also possible to find unsupervised methods, such as the EDFS (Korycki and Krawczyk 2019) that recognize local changes in feature subspaces, they are created in a greedy manner using a round-robin strategy, and using specialized committees on incremental Kolmogorov–Smirnov tests, where a voting scheme is applied among all these tests, ensuring the creation of meaningful feature selections. It has been evaluated on Hyperplane, LED, RBF, SEA, and Tree datasets using Prequential accuracy as the metric chosen, and compared with RFS, lKS, LDCNet, and EHCDD, where the results show that this approach offers significant benefits as compared with univariate approaches, where its performance can be compared with supervised methods. A framework that integrates drift detection and diagnosis (Chapelin et al. 2025) is applied for heterogeneous industrial production processes, leveraging ensemble learning, novelty detection, and continuous learning. This method has been evaluated on private data using accuracy, precision, recall, F1-score, variance, Euclidean and Manhattan distances, and silhouette score, that compared with its different variations mixing KNN, SVM, and LSTM, where the results obtained are promising, but still requiring more optimization to be effective.

The DSE-DD approach (Chambers, Gaber, and Ghomeshi 2025) which can be applied to any DNN, focused on high-dimensional image data. It uses multi-layer DNN activations along with the CBRI descriptor generation technique, as the activation reduction method, to identify concept drift and adapt to it. It has been evaluated on Fashio-MNIST, CIFAR10, and CIFAR100 using average F1-score, average accuracy, and standard deviation, then compared with DDM, ADWIN, HDDMW, KSWIN, MINAS, and OCDD, determining that it outperforms other methods by adding a 5–20% accuracy increase, and reducing the adaptation time nearly 8 times less with 20.5 s and 85% accuracy post-adaptation.

Semi-supervised and hybrid frameworks

Between the semi-supervised approaches, a method based on a distributed Genetic Programming algorithm able to cope with multiple drifts (Folino, Guarascio, and Papuzzo 2019) that utilizes a very efficient detection function based on fractal dimension along with a GP-based decision tree, that generates a new set of labeled tuples when drift is detected, and used it to train new classifiers that will be added to the ensemble, removing the oldest classifiers, so the overall number remains the same. This method has been evaluated on Sine, Hyperplane, Line, Plane, and Circle datasets using Accuracy, Error rate, miss detection rate, and false alarm rate, that after being compared with ADWIN, DDM, and STEPD, it has been determined that it is comparable and also better of the other population-based approaches, but slower than RF approaches. The OPF classifier (Iwashita et al. 2019) utilizes a graph partition task, where each node is encoded by a sample that is connected to others using a predefined adjacency relation by measuring the path-cost using any smooth path-cost function. During the training, the algorithm builds the OPF that is later used in the training to conquer the specific nodes, providing the label of its conqueror with the updated cost. This method has been evaluated on Hyperplane, Usenet, SEA, Covertype, elec2, and Poker hand using Accuracy as the chosen metric to compare with other OPF versions, such as OPF-nomemory, OPF-fullmemory, and OPF-window3, showing that OPF ensemble is suitable compared to traditional OPF methods, where the ensemble showed higher accuracy.

When referring to meta-learning solutions, the main goal is to produce models that have been trained in a wide number of learning tasks, so when they are utilized with new tasks, they can learn with fewer training samples. Kim and Jeon (Kim and Jeon 2025) proposed an approach that uses the logic outputs from pre-trained networks and detects data drift by leveraging internal features extracted by vision models, such as ResNet and Vision Transformers (ViT) (Dosovitskiy et al. 2021), instead of from the raw data. It has been evaluated on ImageNet-1k using accuracy as the metric used to compare the results with other ResNet (50, 101, 152) and ViT (small, base, large), where the results reflect that this lightweight implementation can effectively identify drifts with high accuracy.

Autoencoder and representation-learning methods

In addition, other different strategies and methods have been covered in this study, such as the use of autoencoders, small neural networks that are used to reconstruct input data and capable of learning compact nonlinear codes that summarize the most important features. An unsupervised approach (Jaworski, Rutkowski, and Angelov 2020) demonstrates that by making use of these patterns from one part of the data stream, it can be compared in the incoming parts of the data stream, and monitoring for possible changes in the cost functions, such as the cross-entropy and the reconstruction error, identifying the drift and proceeding with the retraining of the model. It has been evaluated on synthetic datasets using Reconstruction error, and Cross-entropy then the results are compared with RBM being able to demonstrate that autoencoders can be used as detectors. Similarly, the AECDD approach (Kaminskyi, Li, and Müller 2022) employs the LSTM autoencoder, learning the parameters to represent the historical data, and during the detection phase, the reconstruction error is measured and compared with the historical using the Kolmogorov test, determining that the drift is happening. This approach has been evaluated on the SINE dataset using F1-score, accuracy, precision, and reconstruction error, where the results were compared with D3, HDDDM, and CDBD demonstrating it achieves better results even when operating with a non-optimal training size but needs to be extended to multiple drifts.

Also, the VAE-CLT approach (Friedrich, Sawabe, and Hein 2023) aims to identify abnormal behaviors in older adults using data obtained from different sensors, implements a combination of Variational Autoencoders (VAEs) and the Central Limit Theorem (CLT), along with the Kullback–Leibler divergence between the daily activity map from sensors, and the baseline map generated from normal activity patterns. It has been evaluated on Synthetic dataset and data generated with sensors on 20 participant apartments using mean, variance, F1-score, Accuracy, Precision, and Recall, without being compared with other methods, their results are interesting, where they demonstrated to capture the behavior of humans and detect changes by applying the drift detection methods to their patterns of movement.

Semi-supervised neural and AutoML frameworks

A semi-supervised framework designed to detect concept drifts that adversely affect classification performance, known as real drifts, while ignoring those that do not and termed virtual drifts (Castellani, Schmitt,

and Hammer 2021), this approach involves training a neural network to learn a low-dimensional embedding of input data, constrained to ensure that samples from the same class are closely clustered, and those from different classes are well-separated. During inference, the system monitors the distances between incoming data points and these class centroids in the embedding space. Significant deviations from established patterns signal a potential concept drift. This method has been evaluated on RBF and MovingRBF, Adult, bank, digits08, digits17, musk, phishing, and wine datasets using detection accuracy, detection delay, accuracy, and H-score, then compared with HDDDM, IKS, EMAD, ZSD versions with and without the proposed embedding framework, to determine that it effectively identifies drifts impacting the model's predictive accuracy without requiring access to ground truth labels during deployment.

The NRSB (Guo et al. 2023) aims to reduce the negative effect of noise samples near category boundaries, introducing extension and intension distances to define the soft boundaries, and effectively segmenting the feature space. The method implements different labeling strategies depending on the new samples, where automatic labeling is applied to in-class regions, active expert labeling is applied in ambiguous mixed-class regions, and collaborative labeling for novel concepts. It has been evaluated on Hayes-Roth, Page blocks, Yeast, Poker hand, Forest, and KDDCrup99 using Novel precision rate, and Accuracy, then compared with AdBoostM2, Bag, TotalBoost, and MC_SVMA to determine its superior performance, only some experiments identified MC_SVMA as the best. An AutoML study on how to enhance these systems for evolving data (Celik and Vanschoren 2021) proposes six adaptation strategies tailored to detect and respond to various types of concept drift, thereby maintaining model performance over time, and improving the resilience of AutoML systems to data evolution. They have been evaluated on Airlines, Elec2, IMDB, Vehicle, SEA, and Hyperplane datasets using accuracy as the chosen metric to compare the results with Oza bagging and Blast, determining that these strategies proposed allow the autoML to detect, adapt, and recover from drifts successfully.

Key insights

The approaches discussed in this section, summarized in Table 2, demonstrate the increasing role of probabilistic reasoning and ensemble-based adaptation in handling concept drift. Bayesian windowing methods such as SABeDM provide explicit representations of uncertainty, which is particularly useful when observations are sparse or imbalanced. Multi-level schemes such as MWDDM introduce structured warning and drift states that support smoother transitions between models. Ensemble techniques such as SEOA reveal the importance of selective ensemble adaptation for handling concept drift. By dynamically maintaining and updating an ensemble of classifiers according to their predictive performance under evolving data distributions, SEOA is able to effectively accommodate different types of concept drift while preserving robustness and generalization. These developments indicate that modern concept drift handling increasingly relies on adaptive ensemble management and continuous online learning to maintain reliable performance in non-stationary environments.

Catastrophic forgetting

This challenge refers to the phenomenon where a model or system abruptly loses previously acquired knowledge or skills when learning new tasks or adapting to changing environments. This typically occurs because the evolving process updates or replaces parts of the model, often optimizing for recent performance without preserving earlier adaptations. As a result, knowledge accumulated during previous evolving stages can rapidly degrade or disappear entirely, significantly impacting the model's ability to perform effectively on past tasks. This issue stems from the stability-plasticity trade-off: a model must be plastic enough to integrate new patterns but stable enough to prevent the overwriting of existing ones.

A foundational review providing critical context for data-evolving scenarios (Parisi et al. 2019) introduced a taxonomy in the extended study on catastrophic forgetting for deep learning covering more than 200 approaches. Similarly, De Lange et al. (2022) provide an extensive taxonomy, and an overview of different methods, determining the crucial stability-plasticity trade-off, and a comprehensive experimental comparison of 11 of them across multiple benchmarks. Another study focusing on Class-Incremental Learning (CIL) (Masana et al. 2023) a scenario where the model must discriminate between all past and present classes without a task ID during inference. The paper provides a complete survey of existing CIL

methods for image classification and conducts an extensive experimental evaluation on 13 methods across various large-scale datasets and domain shift scenarios. Another article (Aleixo et al. 2024) reviews studies that tackle catastrophic forgetting in deep learning models that rely on gradient descent, providing a new taxonomy, and identify gaps in the area. Teng et al. (2025) systematically investigates different methods, focusing on deeply analyzing their theoretical foundations, specific implementations, and their pros and cons with the aim of producing a classification of them, where another review (Jiang, Fan, and Li 2025) with the focus on visual tasks, where the authors propose a refined classification, and provides a deep analysis bases on their design principles, computational efficiency, and adaptability. Aiming to bridge the gap between basic settings, theoretical foundations, representative methods, and practical applications, the extensive article (L. Wang et al. 2024) summarizes methods with proper stabillity-plasticity trade-off, offering a detailed taxonomy and discussing promising future directions. A review that focuses specifically on the challenges of applying continual learning to Large Language Models (LLMs), where catastrophic forgetting remains a predominant problem after fine-tuning. It structures the field into two directions of continuity: vertical continuity (general to specific capabilities) and horizontal continuity (across time and domains). It reviews methods across Continual Pre-Training (CPT) and Continual Fine-Tuning (CFT) and highlights the need for new evaluation benchmarks for LLMs (Shi et al. 2025). On the other hand, another work argues that forgetting can be beneficial (Z. Wang et al. 2025), for privacy and storage purposes, exploring the phenomenon in non-continual learning contexts. Understanding "why" networks forget is a prerequisite for developing robust mitigation strategies. Recent theoretical research has moved beyond empirical observation to analyze the internal dynamics of neural networks, where Yin et al. (2023) proposed the "Forward Explanation" framework, attributing forgetting to the convergence of task representations and the interference of interleaved representations during training. A structural analysis (Ramasesh, Reibman, and Cox 2020) revealed that deeper layers are significantly more susceptible to forgetting than early layers, and the severity of forgetting is directly correlated with the semantic similarity between tasks. In addition, Mirzadeh et al. (2020) demonstrated that multitask and sequential learning solutions are often connected by linear paths of low error, suggesting that if a model remains within these "stable regions," forgetting can be minimized. Most recently, the field has sought deeper theoretical grounding, with (B. Li, Wang, and Liu 2025) investigating the underlying dynamics of forgetting specifically within two-layer convolutional neural networks, offering mathematical insights into feature evolution and interference. Table 3 provides a summary.

Regularization-based methods

These techniques introduce constrains to the loss function to protect parameters critical for old tasks. Between the weight importance category, Elastic Weight Consolidation (EWC) solution proposes straining important parameters to stay closer to their original values by implementing a function that proportionally decreases their value based on their importance on performance from the previous task, and weighted by utilizing different functions, along with a constraint to them that uses a quadratic penalty. An example that utilizes the Fisher Information Matrix to identify and protect critical weights (Kirkpatrick et al. 2017). Synaptic Intelligence (SI) methods are built on the theoretical idea of synaptic consolidation in neuroscience (Zenke, Poole, and Ganguli 2017), calculating the importance by tracking a synapse's contribution to the reduction of the loss function. An analysis (Nguyen et al. 2019) aimed to integrate developments in task space modeling and correlation analysis, by implementing Task2Vec methods to estimate the actual measures, where they introduced the notion of Total Complexity and Sequential Heterogeneity of task sequences, that it is combined with the correlation analysis by integrating Pearson correlation coefficients. This has been evaluated on MNIST and CIFAR datasets using Average Error Rate, Total Complexity, Sequential Heterogeneity, and Pearson Correlation, where the results were compared with SI, and VCL.

Parameter adaptation

These adaptation methods analyze the weights and biases obtained during the inference constantly, adapting the parameters to mitigate the challenges when they are detected, without involving the alteration of the structural topology group where parameters are changed as a method to mitigate the challenge.

Some of the initial attempts to update parameters used as a solution for catastrophic forgetting by implementing a two-network architecture, where the new items are learned by the first network

Table 3. Summary of catastrophic forgetting methods.

Ref	Category	Method	Datasets
Ashfahani and Pratama (2019)	Structure Adaptation	Autonomous Deep Learning (ADL)	MNIST, Weather, KDDCup99, SEA, Hyperplane, SUSY, Hepmass, RLCPS, RFID Localization
Buzzega et al. (2021)	Parameter Adaptation	IBA, BIC, ELRD, BRS, and LARS	CIFAR-10, CIFAR-100, Fashion-MNIST
Iman et al. 2023)	Structure Adaptation	EXPANSE	MNIST
Nguyen et al. (2019)	Elastic Weight Consolidation	Analysis of Task Sequences and Catastrophic Forgetting	MNIST, CIFAR
Park and Suh (2022)	Parameter Adaptation	Speculative Backpropagation (SB) and Activation History (AH)	MNIST
Ren et al. (2023)	Structure Adaptation	T-SaS	Bouncing Ball, 3-Mode System, Dancing Bees, Elec2, Traffic
Rusu et al. (2022)	Structure Adaptation	Progressive Neural Networks	Multiple Task Domains, Pong Soup, Atari Games, Labyrinth
Soares and Minku (2025)	Parameter Adaptation	Online Weight Averaging (OWA)	Sine1, Sine2, Agrawal3, SEA1, SEA2, STAGGER1, STAGGER2, Elec2, NOAA, Chess, Keystroke, Luxemburg, Ozone, Power Supply, Sensors 29, Sensors 31
Wen et al. (2020)	Structure Adaptation	AutoGrow	CIFAR10, CIFAR100, SVHN, FashionMNIST, MNIST, ImageNet
Yoon et al. (2018)	Structure Adaptation	Dynamically Expandable Network (DEN)	MNIST-Variation, CIFAR-100, AWA
Zhong et al. (2023)	Structure Adaptation	Dynamically Evolving Deep Neural Network (DeDNN)	Airfoil, WineQ, Rate1, ETTh1, CaliF, Bitcoin, Rate2, AirQ, CondM, SP500, OilF, Rate, HouseEP
Schwartz et al. (2020)	Key Insights	N/A	N/A
Teng et al. (2025)	Other	Theoretical analysis ans comparison	N/A
Jiang, Fan, and Li 2025)	Other	Analysis on Visual approaches	N/A
L. Wang et al. (2024)	Other	Summary of methods based on plasticity-elasticity	N/A
Shi et al. (2025)	Other	Continual Learning on Large Language Models	N/A
Z. Wang et al. (2025)	Other	Catastrophic Forgetting for Privacy and Storage	N/A
Yin et al. (2023)	Other	Analysis of Internal Dynamics of Neural Networks	N/A
Ramasesh, Reibman, and Cox (2020)	Other	Structural Analysis of Catastrophic Forgetting	N/A
Mirzadeh et al. (2020)	Other	Stable Regions in Continual Learning	N/A
Kirkpatrick et al. 2017)	Regularization-based	Elastic Weight Consolidation (EWC)	N/A
Zenke, Poole, and Ganguli (2017)	Regularization-based	Synaptic Intelligence (SI)	N/A
Lee et al. (2017)	Parameter Adaptation	Variational Continual Learning (VCL)	N/A
Lopez-Paz and Ranzato (2017)	Replay and Memory-Based	Gradient Episodic Memory (GEM)	N/A
Riemer et al. (2019)	Replay and Memory-Based	Meta Experience Replay (MER)	N/A
Wu et al. (2019)	Replay and Memory-Based	Bias Correction (BiC)	N/A
Rebuffi et al. (2017)	Replay and Memory-Based	Incremental Classifier and Representation Learning (iCaRL)	CIFAR-100, ImageNet ILSVRC
Serra et al. (2018)	Parameter Adaptation	Hard Attention to the Task (HAT)	N/A
Tadros et al. (2022)	Replay and Memory-Based	Sleep-like Replay	N/A
Carta et al. (2022)	Replay and Memory-Based	Theorical analysis of Replay Methods for Graph Neural Networks	N/A
B. Li, Wang, and Liu (2025)	Other	Analysis of methods	N/A

concurrently with internal pseudo-items generated from the second network, reflecting the structure of the previously learned network, where it was experienced the possibility of selectively slowing down the learning on the weights for new tasks that are important and protecting the important weights of the old previously learned tasks, kind of humans when they expend long periods learning and perfecting skills, where an study proposed two different approaches (Park and Suh 2022), a Speculative Backpropagation method (SB) that aims to readjust the weights by calculating derivates of the error to replicate back the past knowledge, and a Biased weight update with Activation history (AH), where they were able to preserve the important weights from a previous task and confirm that there are two types of activated neurons behaviours, (a) While training one specific task, the neurons activated in the past are more likely to be activated in the future, and the deactivated neurons are more likely to be deactivated. (b) The activated neurons differ from task to task. Only activated neurons affect the inference result and play an important role in that task. This has been evaluated on MNIST dataset using Accuracy as the metric to compare with Gradient descent, Joint training, EWC, and SI using an MLP as baseline, where the results show that continual learning ability is improved when using SB and AH. Incremental Moment Matching (IMM) (Lee et al. 2017) aims to find a parameter space that satisfies the posterior distributions of both old and new tasks by matching their statistical moments.

On the other hand, an attempt to mimic the human brain for sequential learning using a multilayer model following Occam's razor principle, consisting of interleaving past examples with the current training batch has been extended with the proposal of five tricks to mitigate the shortcomings that restrain Experience Replay (Buzzega et al. 2021), covering Independent Buffer Augmentation (IBA), Bias Control (BIC), Exponential Learning Rate Decay (ELRD), Balanced Reservoir Sampling (BRS), and the Loss-Aware Reservoir Sampling (LARS). They have been evaluated on CIFAR-10, CIFAR-100, and Fashion-MNIST using Average accuracy, and execution times as metrics to compare with SGD, Joint training, A-GEM, GEM, HAL, iCaRL, and ER, where the results displayed that ER along with the tricks outperforms more sophisticated approaches. Moving toward architectural constraints, Serra et al. (2018) introduced Hard Attention to the Task (HAT), utilizing learnable attention masks to preserve task-specific parameters without affecting future learning.

In general, methods focus on the forgetting nature of the models, where Soares and Minku (2025) proposed the Online Weight Averaging (OWA) approach based on stochastic weight averaging, implementing an ensemble of neural networks that work as a single model, and combining snapshots of different time steps and utilize those weights by averaging them and update the model. This method has been evaluated on Sine1, Sine2, Agrawal3, SEA1, SEA2, STAGGER1, STAGGER2, Elec2, NOAA, chess, keystroke, Luxemburg dataset, Ozone, Power supply, Sensors 29 and Sensors 31 datasets using G-mean, and prequential accuracy as metrics to compare with Ozabag, Ozabag-adwin, DWM, CSB2, SRPC, KUE, CALMID, MICROFAL, and SWA, where the results displayed that OWA clearly outperforms SWA, and deliver similar or better predictive performance than other ensemble methods.

Structure adaptation

The structure adaptation methods aim for the modification of the network topologies, which can be done by adding or removing neurons or layers based on the new data requirements.

The Growing Neural Networks solutions aim to increase the number of neurons or layers on the topology, allowing the incremental addition of neurons and connections between them, helping to adapt to changes in the distribution of the data.

As an example, Zhong et al. (2023) propose a dynamic online learning model called Dynamically evolving Deep Neural Network (DeDNN), capable of structural self-adjustment in response to non-stationary data streams. DeDNN evolves from a basic multilayer perceptron, adding neurons based on local prediction errors and adding hidden layers guided by changes detected using Jensen-Shannon divergence. An adaptive memory mechanism preserves both local and global samples to avoid catastrophic forgetting as the network structure dynamically expands. This method has been evaluated on Airfoil, WineQ, Rate1, ETTh1, CaliF, Bitcoin, Rate2, AirQ, CondM, SP500, OilF, Rate and HouseEP datasets using RMSE, and total running time as the metrics of choice to compare with NADINE, where the results showed clear superiority.

The dynamic NAS techniques automate the design of neural networks by enabling the models to grow, shrink, or modify their architecture dynamically during running time, and without forgetting previous knowledge, such as the DEN method (Yoon et al. 2018) uses an efficient way of using group sparse regularization to dynamically decide the number of neurons to add at which layer, avoiding fully retraining the model, this is done by checking the loss is below a certain threshold, otherwise, it will expand its capacity and the hidden units that are deemed unnecessary from the training will be dropped altogether. The evaluation on MNIST-Variation, CIFAR-100, and AWA using Accuracy, AUGROC, capacity, and expansion performance, where the results have been compared with DNN-MTL, DNN, DNN-L2, DNN-EWC, and DNN-Progressive, helping to determine that DEN significantly outperform the existing methods. This is extended in the ADL approach (Ashfahani and Pratama 2019) with a flexible structure, that can be built from scratch using a construct, and utilizes the Network Significance formula to drive the hidden nodes growing and pruning mechanisms. This method has been evaluated on MNIST, Weather, KDDCup99, SEA, Hyperplane, SUSY, Hepmass, RLCPS, and RFID localization data using classification rate, execution time (ET), HL, HN, and the number of parameters (NoP) as the metrics to compare with pE+, pE, DFN, and DNN, showing that it consistently outperforms the compared methods. Contrary to these methods, EXPANSE (Iman et al. 2023) proposed expanding pre-trained deep learning models vertically, adding neurons to the existing layers rather than horizontally as other approaches suggest. It has been evaluated on MNIST dataset using Accuracy, and compared to different version of DNN, where the results demonstrated the effectiveness of using two-step training, and increasing the whole model's learning capacity through vertical expansion of the model. The AutoGrow (Wen et al. 2020), a method that automates the depth discovery on Deep Neural Networks, beginning with a shallow seed structure, it adds new layers if it will improve the accuracy of the model until it cannot be further improved. This method has been evaluated on CIFAR10, CIFAR100, SVHN, FashionMNIST, MNIST, and ImageNet using metrics such as Accuracy, difference between accuracies in autogrow after and before training, and growing time, where the results were compared with different implementation of ResNet to conclude that this rapid growing outperforms more intuitively correct growing methods. Similarly, the introduction of Progressive Neural Networks (Rusu et al. 2022) addresses the catastrophic forgetting challenge presented in Neural Networks by expanding the network with new neurons and lateral connections in charge of the transfer learning from older tasks, without modifying existing parameters. Between the current limitations of this kind of method, the growth occurs indefinitely with each new task, requiring manual intervention to include a new column, where a future optimization will be required to provide the model with capabilities to avoid the creation of expansions while new tasks are like previous ones. This approach has been evaluated on Sample data from different task domains related games using ROC AUC, and transfer score metrics to compare with a baseline Reinforcement Learning model, where the results displayed that this method is immune to forgetting. The T-SaS method (Ren et al. 2023) is based on a Bayesian framework with a discrete distribution-modeling variable that helps to capture sudden distribution changes and adapt dynamically to them by selectively activating subsets of neurons. This method has been evaluated on Bouncing ball, 3 mode system, Dancing bees, elec2, and Traffic datasets using Accuracy, Normalized Mutual Information, and Adjusted Rand Index as metrics to compare with MOCA, AdaRNN, ARIMA, DeepAR, ARSGLS, KVAE-RB, and JT, where the results displayed a superior performance in detecting and adapting.

Replay and memory-based

Replay methods address catastrophic forgetting by mimicking biological memory consolidation, specifically the mechanism of revisiting previous experiences while learning new tasks. These approaches typically maintain a small buffer of data from prior tasks to regularize the training of the current task.

Early approaches focused on using this stored data to constrain optimization. For instance, Gradient Episodic Memory (GEM) (Lopez-Paz and Ranzato 2017) utilizes a small subset of episodic memory to guide gradient updates. Rather than simply mixing old and new data, GEM treats the stored samples as inequality constraints, ensuring that updates for the new task do not increase the loss on previously learned tasks. Rebuffi et al. (2017) established a seminal baseline with iCaRL, proposing an incremental classifier and representation learning framework that combines nearest-neighbor classification with exemplar-based rehearsal and knowledge distillation; iCaRL selects and stores a small subset of "exemplar" images that

best represent the class mean for previous tasks. During training, it combines these stored images with new data and uses a distillation loss to ensure the network's predictions on old classes do not drift significantly.

Building on the concept of gradient interaction, Meta-Experience Replay (MER) (Riemer et al. 2019) integrates meta-learning into the replay framework. While GEM focuses on preventing interference (constraints), MER explicitly aims for gradient alignment. It optimizes parameters such that gradients from new and old tasks are aligned, thereby maximizing knowledge transfer while simultaneously minimizing interference.

However, the use of replay buffers can introduce unintended artifacts in large-scale incremental learning, such as "recency bias" – a tendency for the model to favor new classes over old ones. To mitigate this, Wu et al. (2019) introduced Bias Correction (BiC). This method applies a dedicated bias correction layer to rescale the output logits, effectively balancing the classifier's predictions between old and new classes to ensure fair evaluation across all learned tasks.

Recent work has expanded these paradigms into biological and non-Euclidean domains (Tadros et al. 2022); demonstrated the efficacy of sleep-like unsupervised replay, mimicking biological memory consolidation to reduce forgetting.

The work presented in the study (Carta et al. 2022) is not a method per se but a foundational study establishing that Graph Neural Networks (GNNs) suffer heavily from forgetting. It benchmarks existing strategies (like LwF and EWC) on graph topology data, finding that Replay is often the only effective strategy for graphs, while regularization methods often fail.

Key insights

The methods surveyed here reflect a growing emphasis on adaptive model capacity as a means of mitigating catastrophic forgetting. Approaches such as DEN (Yoon et al. 2018) adopt sparsity-based strategies that encourage the reuse of existing representations while expanding capacity only when necessary, thereby supporting efficient and resource aware adaptation. This is all the more relevant when considering more efficient and greener architectures, as a higher proportion of frozen parameters directly reduces the amount of training required (Schwartz et al. 2020). Techniques such as AutoGrow (Wen et al. 2020) illustrate the potential of depth expansion, although they also underscore the computational and latency constraints associated with increasingly deep architectures. These considerations are especially important in continuous stream settings in which task boundaries are not explicitly defined. The overarching insight is that model adaptation in EML requires mechanisms that balance stability and plasticity while preserving computational viability in real-time environments.

Skewed learning

This challenge refers to the difficulty of learning from data streams where the distribution of classes is highly imbalanced, and certain classes may become significantly underrepresented, leading to poor generalization and biased predictions.

Between the surveys visited for this challenge, Johnson and Khoshgoftaar (2019) surveyed 131 methods for deep learning with class imbalance. Untoro et al. (2020) provided a survey and comparison of 23 base classifiers, such as Decision Tree, K-NN, Naïve Bayes, and SVM methods on the UCI dataset, and Kumar et al. (2021) present a review of methods and their applications with a total of 25 research papers.

The resampling techniques aim to modify the distribution or size of data samples to improve performance in dynamic or imbalanced environments, where GABagging (Y. Wang 2019) utilizes a sample combination optimization genetic algorithm that reduces part of the majority of the class samples, while maintaining the number of classes, creating new sets and combining them until a subset is reached with balanced samples, which then is used to train several classifiers. It has been evaluated on b1, breast, yeast, wisconsin, vehicle, segment, pima, haberman, glass, and spect datasets using TPR, and AUC, then compared with OverBagging, UnderBagging, ABBag, HSBagging, and RBSBagging, where the results showed that this method has good prediction effect, but its time complexity is too high. Similarly, Aminian, Ribeiro, and Gama (2021) introduced a technique that addresses imbalanced data in regression tasks, and specifically when rare and extreme values are of significance, employing sampling strategies, under-sampling and over-sampling, that are guided by Chebyshev's. This method has been evaluated on 14 benchmark datasets using

Accuracy, where the results proved its superiority. A hybrid approach (Czarnowski 2021) processes incoming data in chunks, applying over-sampling to augment minority class instances and instance selection techniques to refine the dataset. This balanced data chunk is then used to train ensemble classifiers. The approach has been evaluated using benchmark datasets using F-measure, and G-mean, where the results show an improved performance of baseline classifiers.

On the other side, AMSCO (Li et al. 2018) implements two stochastic swarm heuristics optimizers, that cooperatively find a close to optimal mix with high accuracy and reliability. It has been evaluated on Abalone, Cleveland, Flare, glass, Haberman, Pima, Poker hand, vehicle, Wine, and yeast datasets using Accuracy, standard deviation, precision, recall, F1-score, and Kappa as metrics, comparing the results with NN, Bagging, AdaBM1, CTS, DT, AdaC, SBO, RUS, BC, EE, and ADASY, where it was determined that it excellently overcomes other methods. An optimized Random-SMOTE (L. Zhang et al. 2019) that utilizes K-means clustering to identify the centroids for the minority classes and calculate their distance to the centroids of the majority ones, where the samples with the smaller distance are then taken for oversampling using the Random-SMOTE algorithm to finish to balance the data in the data level. Then, the Biased-SVM algorithm is used to balance the data at the classifier level by setting the different penalty factors for the minority and majority samples. It has been evaluated on Banana, Haberman, Appendicitis, Vehicle, and Wisconsin datasets using Accuracy, and F1-score as chosen metrics to compare with SVM, and base Random-SMOTE, where it was determined that this approach adds some improvements on the classification effects, but the optimization of the parameters still requires more optimization.

Under the cost-sensitive learning type of methods, the CMBoost (Z. Zhang, Qiu, and Dai 2019) algorithm is based on the cost-sensitive margin statistical characteristics, prioritizing minority-class recognition, reducing overfitting, and obtaining fewer false negatives, that are combined with a boosting algorithm. This method has been evaluated on heart, sonar, vehicle, wine, wpbc segment, and vote datasets using Recall, precision, Positive accuracy, Negative accuracy, F-measure, and G-mean as the metrics to compare with AdaBoost, where the results showed an increase in performance. On the other side, C-OSELM (Wen Wang, Guo, and Cheng Yan Y 2021) introduced an online sequential extreme learning machine, as the basic theory model, combined with a cost-sensitive strategy, to improve classification performance for the minority classes. This involves the assignation of different penalty parameters to the different classes, adjusting the misclassification costs and mitigating the decision boundary bias, by incorporating a cost adjustment function that dynamically optimizes the weight parameters. This method has been evaluated on Pima, yeast, Haberman, vehicle, segment, yeast Ecoli, page-blocks, Vowel, Led, Shuttle, and Abalone datasets using G-mean as the metric used to compare the performance against OSELM and WOS-ELM, where the results displayed that it was significantly better in almost all the experiments. The CSRDA method (Chen, Sheng, et al. 2023) extends the RDA approach by including a new optimization function based on cost sensitivity that prioritizes the correct classification of minority classes, and enforces sparsity when handling severely imbalanced datasets. This approach has been evaluated on six benchmark datasets using G-mean to compare with baseline methods, showing that it improves classification performance, and captures sparse features efficiently, resulting in better model interpretability.

Between the ensemble methods for this challenge, a hybrid approach that combines the SMOTE over-sampling technique, aiming to generate synthetic samples for the minority class, and the XGBoost algorithm, called SMOTEXGBoost (Mardiansyah, Sembiring, and Efendi 2021), is well-known for its robustness. This method has been evaluated on German, Winsconsin, Glass, and E-coli using Accuracy, and AUC as metrics to compare with SMOTE, XGBoost, SMOTEXGBoost, LR, RF, and SVM, determining that even the results show an increase on the performance, it was proven the lack of stability of the approach. Another SMOTE over-sampling generation in combination with Gaussian Mixture Model (Tripathi et al. 2021) where the synthetic data samples are selected based on the weight associated with each cluster and determined by the distribution of the majority class samples. This method has been evaluated on CM, PC, MC, MW, KC, and Haberman datasets using Precision, Recall, and sensitivity to compare with SMOTE, SMOTE-RUS, SMOTE-Tomek Links, and SMOTE-OSS, where the results concluded that it showed a higher sensitivity than others and performed better. WECOI approach (Czarnowski 2022) integrates a weighted ensemble of one class classifiers with over-sampling and instance selection techniques to balance the class distribution, generating synthetic data samples for the minority classes. It has been evaluated on Heart, Diabetes, WBC, Acredit, Gcredit, Sonar, Satellite, Banana, Image, Thyroid, Spambase, Twonorm,

Table 4. Summary of skewed learning methods.

Ref	Section	Category	Method	Datasets
J. Li et al. (2018)	Resampling	Supervised	Adaptive Multi-objective Swarm Crossover optimisation (AMSCO)	Abalone, Cleveland, Flare, Glass, Haberman, Pima, Poker hand, Vehicle, Wine, Yeast
Y. Wang (2019)	Resampling	Supervised	GABagging	B1, Breast, Yeast, Wisconsin, Vehicle, Segment, Pima, Haberman, Glass, Spect
L. Zhang et al. (2019)	Resampling	Supervised	Optimised Random-SMOTE with Biased-SVM	Banana, Haberman, Appendicitis, Vehicle, Wisconsin
Aminian, Ribeiro, and Gama (2021)	Resampling	Supervised	Under and over sampling using Chebyshev's	14 Benchmark datasets
Czarnowski (2021)	Resampling	Supervised	Hybrid oversampling and instance selection	Benchmark datasets
Z. Zhang, Qiu, and Dai (2019)	Cost-sensitive	Supervised	CMBoost	Heart, Sonar, Vehicle, Wine, WPBC, Segment, Vote
Wen Wang, Guo, and Cheng Yan Y (2021)	Cost-sensitive	Not specified	C-OSELM	Pima, Yeast, Haberman, Vehicle, Segment, Ecoli, Page-blocks, Vowel, Led, Shuttle, Abalone
Chen, Sheng, et al. (2023)	Cost-sensitive	Not specified	Cost-Sensitive Regularized Dual Averaging (CSRDA)	6 Benchmark datasets
Mardiansyah, Sembiring, and Efendi (2021)	Ensemble	Supervised	SMOTEXGBoost	German, Wisconsin, Glass, E-coli
Tripathi et al. (2021)	Ensemble	Supervised	Cluster-based approach combining SMOTE oversampling with Gaussian Mixture Model clustering	CM, PC, MC, MW, KC, Haberman
Czarnowski (2022)	Ensemble	Supervised	Weighted Ensemble with One-class Classification and Oversampling and Instance Selection (WECOI)	Heart, Diabetes, WBC, Acredit, Gcredit, Sonar, Satellite, Banana, Image, Thyroid, Spambase, Twonorm, Elec2, Airlines, Ozone, Gas, Hyperplanes, AGR, SEA

Elec2, Airlines, Ozone, Gas, hyperplanes, AGR, and SEA datasets using Average accuracy to compare with AWE, HOT, iOVFDT, OUOB, OB, Learn++.NIE, and WECU, where the results demonstrated to be a competitive solution for them. Table 4 provides a summary.

Key insights

The studies reviewed in this section show that class imbalance interacts strongly with distributional drift and can significantly alter model behavior. Oversampling techniques such as SMOTE may become unreliable when the minority class shifts over time, since they can amplify outdated regions of the feature space and thereby degrade precision and recall. In contrast, approaches such as C-OSELM offer more stable performance by adjusting decision boundaries without synthesizing new samples, which reduces the risk of reinforcing obsolete structures. These observations emphasize that imbalance handling in EML must be designed with explicit consideration of drift dynamics to ensure robust model adaptation.

Synthesis of cross-cutting insights

Across the methods reviewed in this survey, a coherent view of EML emerges in which the tasks of detecting, interpreting, and adapting to change form an integrated pipeline rather than a sequence of isolated operations. In the context of data drift, approaches such as DetectA, LASSO, and Adaptive SPLL illustrate the progression from reactive error-driven detection toward strategies that monitor input distributions, structural feature importance, and adaptive windowing. These developments emphasize the importance of balancing early sensitivity to change with robustness to noise and gradual evolution.

Within concept drift and ensemble-based adaptation, methods such as SABeDM, MWDDM, SEOA, and OWA demonstrate the growing role of uncertainty modeling, structured warning mechanisms, and both function space and parameter space adjustment. These techniques collectively highlight that modern drift handling must couple reliable detection with principled mechanisms for maintaining stability during transition.

In addressing catastrophic forgetting, approaches including DEN and AutoGrow reveal an increasing emphasis on adaptive model capacity. These methods encourage selective retention of prior knowledge and

introduce new representational resources only when required, while also drawing attention to the computational constraints implicit in stream-based and latency-sensitive environments.

Finally, the analysis of skewed learning shows that imbalance handling interacts fundamentally with drift. Techniques such as SMOTE may reinforce outdated minority structures when class distributions move, whereas cost-sensitive strategies such as C OSELM provide more reliable adaptation by avoiding assumptions about the location of minority examples.

Taken together, these insights indicate that EML is evolving into a unified paradigm in which systems must integrate proactive and uncertainty aware detection, informed interpretation of structural change, and adaptive mechanisms that span functions, parameters, and model capacity. The field increasingly recognizes that effective learning under non-stationary conditions requires coordinated design across these components rather than reliance on any single methodological perspective. The challenges discussed above, while presented separately for clarity, are rarely encountered in isolation in real-world evolving environments. This motivates a consolidated analytical perspective that connects their interdependencies.

An integrative perspective on evolving machine learning

Although the preceding subsections examine data drift, concept drift, catastrophic forgetting, and skewed learning individually, these phenomena represent interconnected manifestations of non-stationarity within evolving environments. A unified understanding of EML requires analyzing how distributional shifts and model adaptation dynamics interact over time rather than treating these challenges as isolated problems.

From a distributional standpoint, evolving environments may experience changes in different components of the data-generating process. Data drift reflects variations in the input feature distribution, while skewed learning commonly arises from shifts in class distributions, including evolving imbalance or the emergence of new classes. Concept drift corresponds to changes in the relationship between inputs and outputs, directly altering decision boundaries and predictive behavior. In contrast, catastrophic forgetting emerges at the model level when sequential adaptation to new data interferes with previously acquired knowledge.

These dimensions are structurally interdependent. Changes in input distributions may precede or induce modifications in predictive relationships; evolving class distributions may destabilize previously learned decision boundaries; and aggressive adaptation strategies designed to handle drift may inadvertently intensify forgetting. Conversely, strong memory-preserving mechanisms may reduce responsiveness to abrupt distributional changes. As a result, mitigation strategies targeting one challenge can influence the behavior of others.

Within the unifying paradigm adopted in this survey, EML is, therefore, interpreted as a coordinated response to multiple, interacting forms of non-stationarity. Robust evolving systems must simultaneously manage distributional variation, adaptive updating, stability–plasticity trade-offs, and imbalance dynamics. This integrative perspective highlights that sustained learning under dynamic conditions requires balancing responsiveness to new information with the preservation of accumulated knowledge, thereby linking drift management, skew handling, and forgetting mitigation within a coherent analytical view.

Comparative analysis and case studies

Building on the integrative perspective outlined in An integrative perspective on evolving machine learning, we now examine recent methodological advances that operationalize these adaptive mechanisms across evolving settings. This section presents a comprehensive comparative analysis of existing EML solutions and their suitability for real-world applications. It reviews both survey and research papers to identify key trends, benchmark methodologies, and evaluate the maturity of research within the field.

Comparative study of methods

A total of 28 survey papers were reviewed to capture recent developments addressing the primary challenges in EML. Figure 3 illustrates the distribution of these surveys across the identified challenges. The results indicate that Concept Drift is the most extensively studied area,

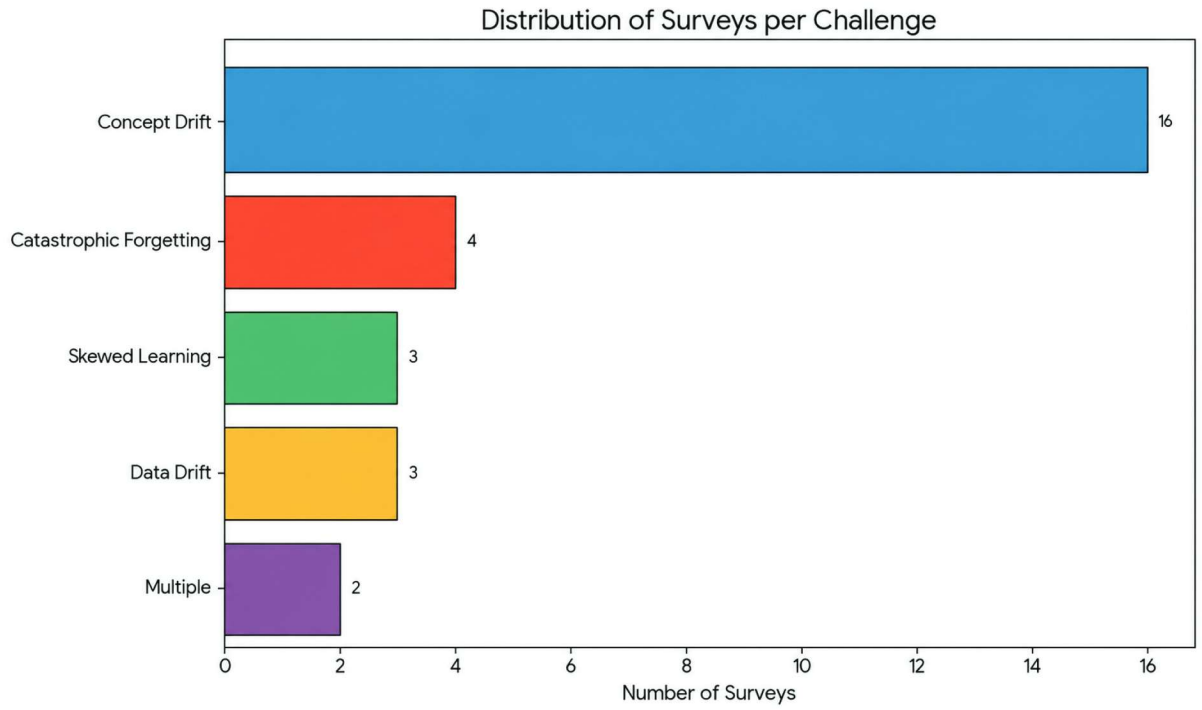

Figure 3. Survey distribution per challenge.

encompassing 16 of the reviewed survey, followed by Catastrophic forgetting with 4 surveys, then Skewed Learning and Data Drift each represented by three surveys. In particular, a single survey covers all the related challenges comprehensively, and the preliminary print version of this survey submitted to arXiv, represented as Multiple with 2.

Figure 4 presents the temporal distribution of survey publications. The majority of surveys reviewed were published in 2023, with six papers, followed by the year 2019 with five papers. Four were published in 2020, 2022, and 2025, while three of them in 2024, and only isolated studies appeared in 2018 and 2021. This consistent publication activity demonstrates that EML remains an active and evolving research area.

Beyond survey papers, this work also reviews three studies related to ethical and policy aspects related to EML, and 94 primary studies addressing the four key EML challenges. Figure 5 depicts their distribution:

- Concept Drift dominates with 45 studies,
- Catastrophic Forgetting with 31.
- Skewed Learning follows with 11,
- Data Drift with 7,

This uneven distribution reflects the differing levels of maturity among the four core EML challenges. Concept drift and catastrophic forgetting have been extensively studied, leading to a substantial body of methodological and empirical research. By comparison, data drift and skewed learning have received less attention despite their practical importance in evolving environments. This imbalance highlights opportunities for future research and further motivates the need for a unified EML perspective that considers these challenges collectively rather than in isolation.

Data drift

Analysis of the 7 data-drift studies, see Figure 6, shows that two methods addressed all drift types, while one each focused on abrupt and simultaneous drifts; three did not specify a drift type. Benchmark datasets include synthetic sources (Sine, Circles, Stagger, Hyperplane, SEA) and real-world datasets (Elec2,

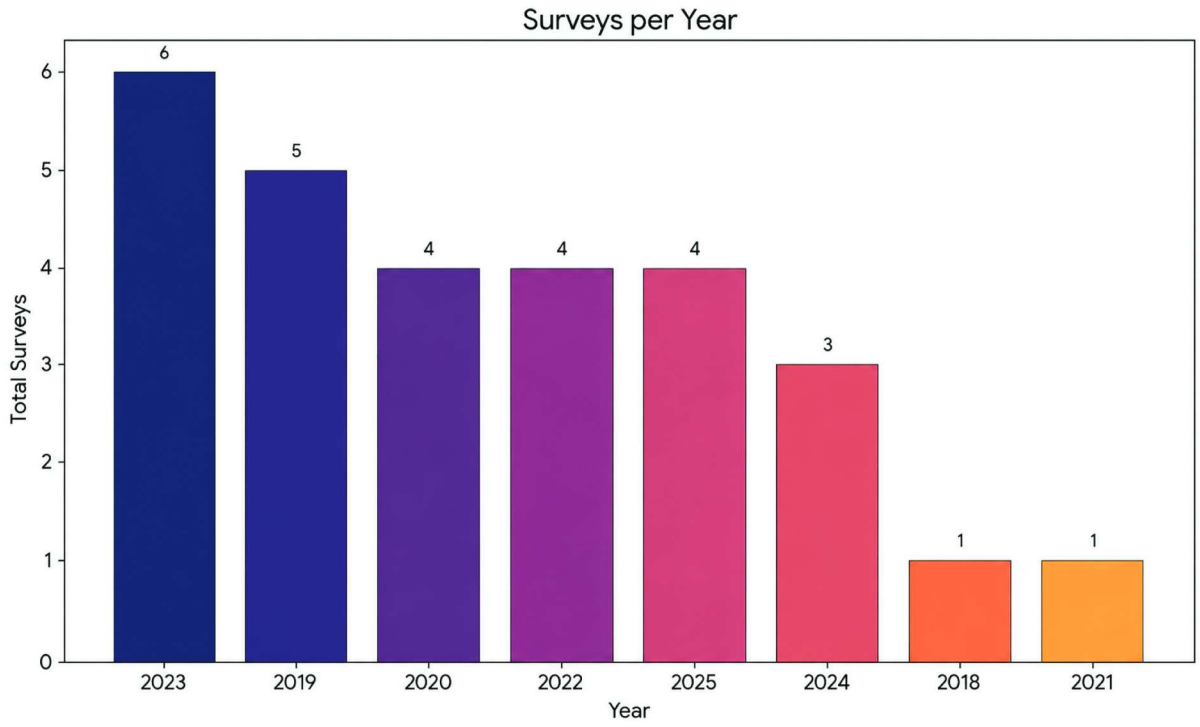

Figure 4. Survey distribution by year.

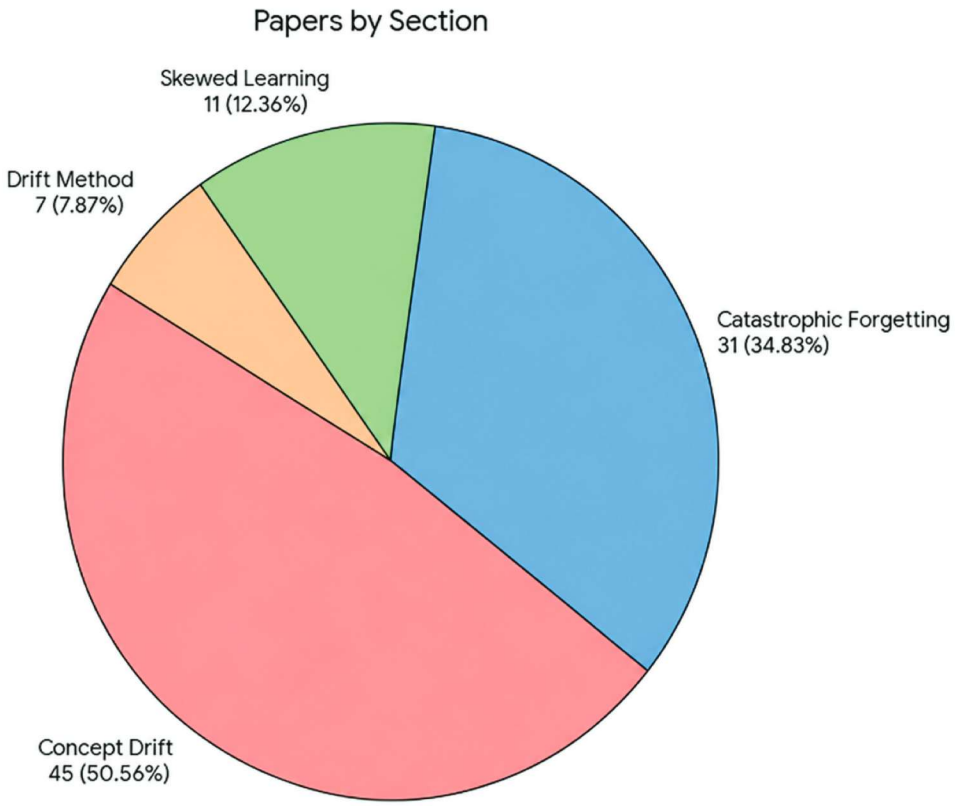

Figure 5. Distributions by challenge.

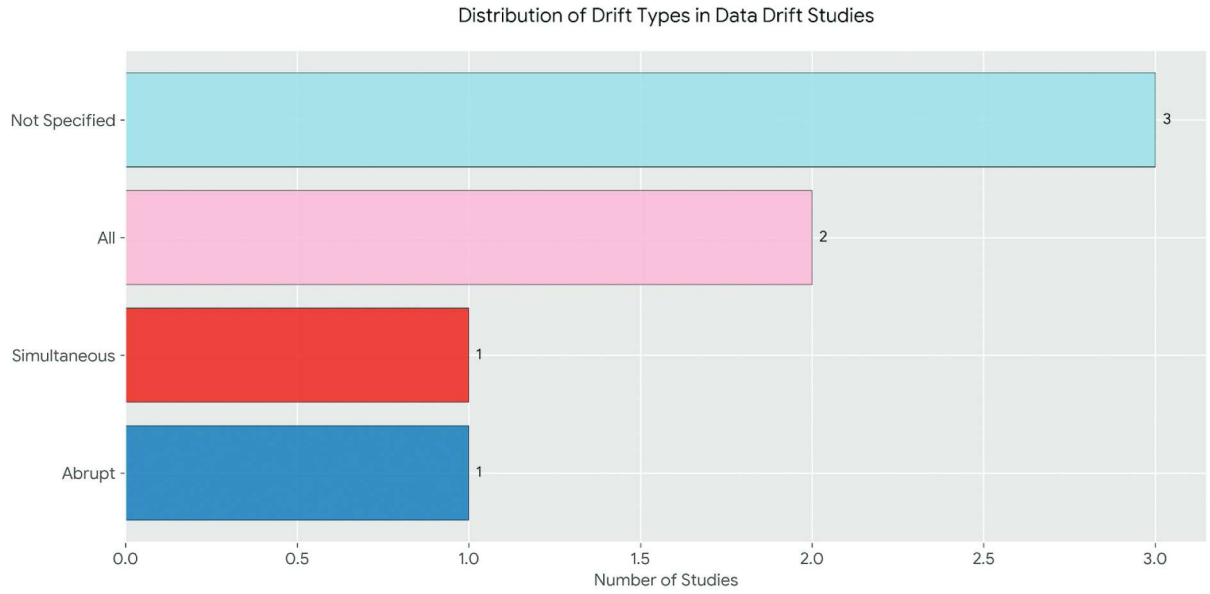

Figure 6. Drift type distribution on data drift approaches.

CoverType, Poker Hand, Rialto, Airlines, CIFAR10). Baseline comparison methods primarily involve DDM and EDDM, evaluated with metrics such as magnitude, accuracy, and prequential accuracy.

The comparatively small number of data-drift studies identified in this review contrasts sharply with the extensive literature on concept drift. This observation suggests that research efforts have predominantly focused on changes in the input–output relationship, while shifts in the input distribution itself remain relatively underrepresented despite their importance in real-world deployments.

Concept drift

Among the 45 concept drift studies, Figure 7, a variety of drift types were reported. Even if most of the methods do not specify the type of drifts with 17 studies, the "Abrupt and gradual" and "various" category are the most common, showing six studies each, followed by "all" with 3, then "Sudden" and "Sudden, gradual" with 2 each, and several combinations of different drifts are present with 1 study each.

As shown in Figure 8, sliding-window-based methods dominate this area (24 studies), followed by ensemble-based (13), other (6), and meta-learning approaches (2). Benchmarks often use synthetic datasets (Sine, Agrawal, SEA, Stagger, Hyperplane) and real-world datasets (Airlines, Elec2, MNIST, CIFAR, CoverType, Poker Hand, Rialto, KDDCup99). Common baselines include DDM, EDDM, ADWIN, HDDMW, SMOTE, STEPD, D3, and NEAT, with evaluation metrics such as accuracy, F1-score, G-mean, detection time, and number of drifts.

Catastrophic forgetting

The Catastrophic Forgetting challenge has been addressed by 31 main studies, see Figure 9. Overall, the graph indicates that papers are concentrated in broader or structural adaptation categories, while more specific regularization-based methods, particularly Elastic Weight Consolidation, are comparatively rare. The most frequent approaches are those refereed as others with nine papers, that contains theoretical, empirical and different analysis, followed by structure adaptations with seven papers, Replay and memory-based methods with six papers, Parameter adaptation includes five papers, Regularization-based are less common with two papers, and the least represented is Elastic Weight Consolidation with one paper, but also mention an study that helped draw the key insights. Benchmark datasets are primarily synthetic (SEA, Hyperplane) or standard real-world image datasets (MNIST, CIFAR variants). Unlike other sections, no common baseline exists, as each study employs specialized deep neural network (DNN) comparison models, evaluated primarily by accuracy and error rate.

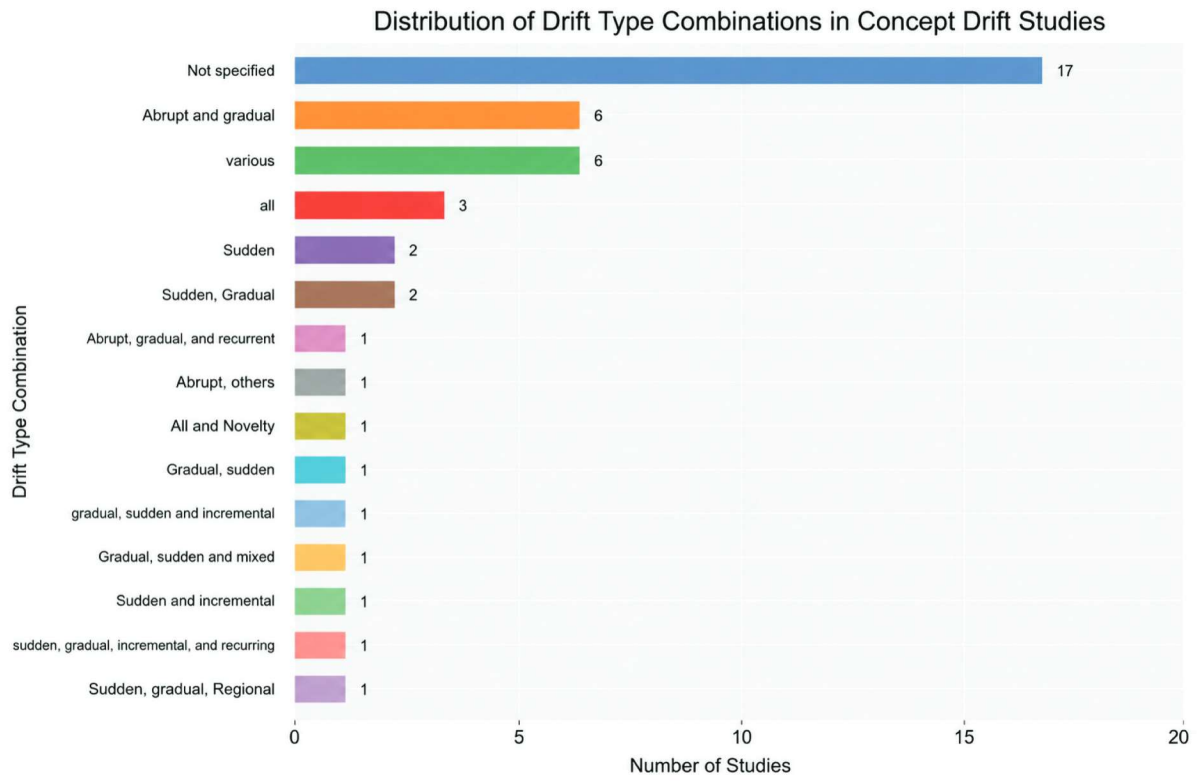

Figure 7. Distribution based on drift type.

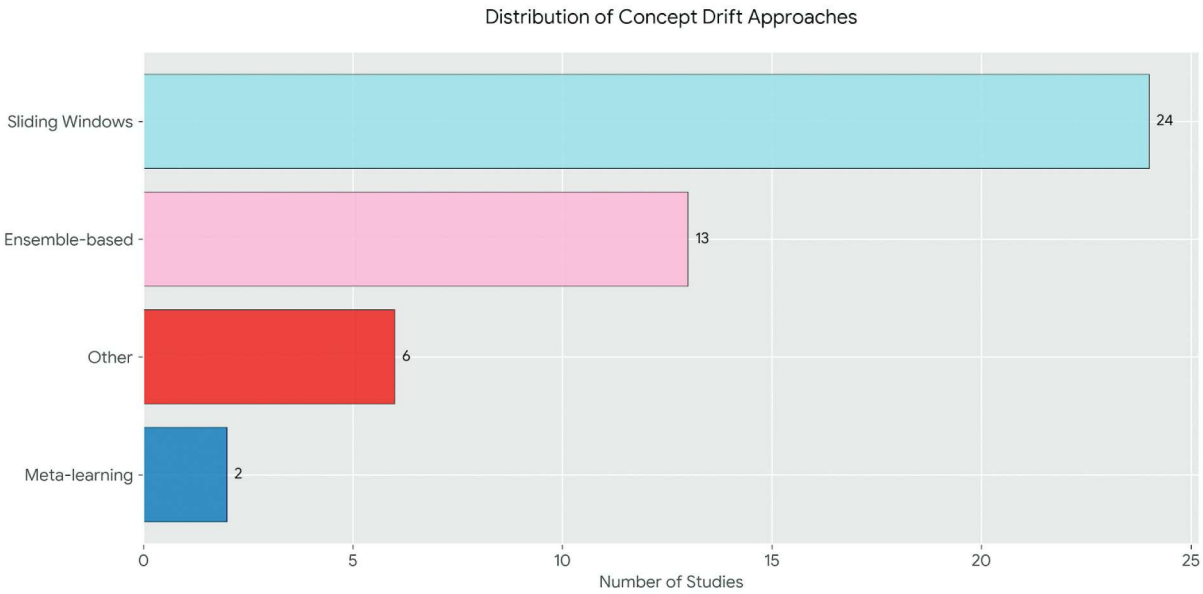

Figure 8. Distribution based on the type of approach.

Skewed learning

The Skewed Learning category includes 11 studies, Figure 10. Most employ resampling techniques (five studies), followed by cost-sensitive and ensemble methods (three each). Common datasets include Yeast, Wisconsin, Vehicle, E-coli, Sonar, and Glass. Baseline models frequently use bagging and SMOTE variants, with performance measured via accuracy, G-mean, and F-measure.

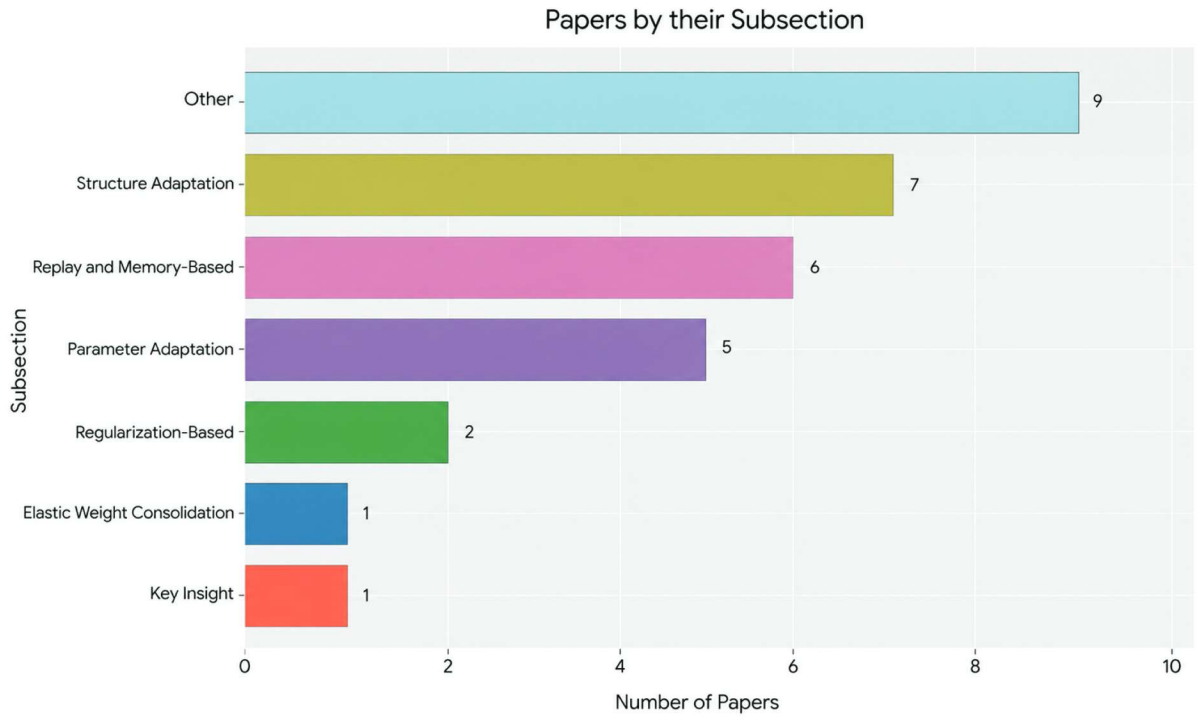

Figure 9. Distribution of approaches based on the section.

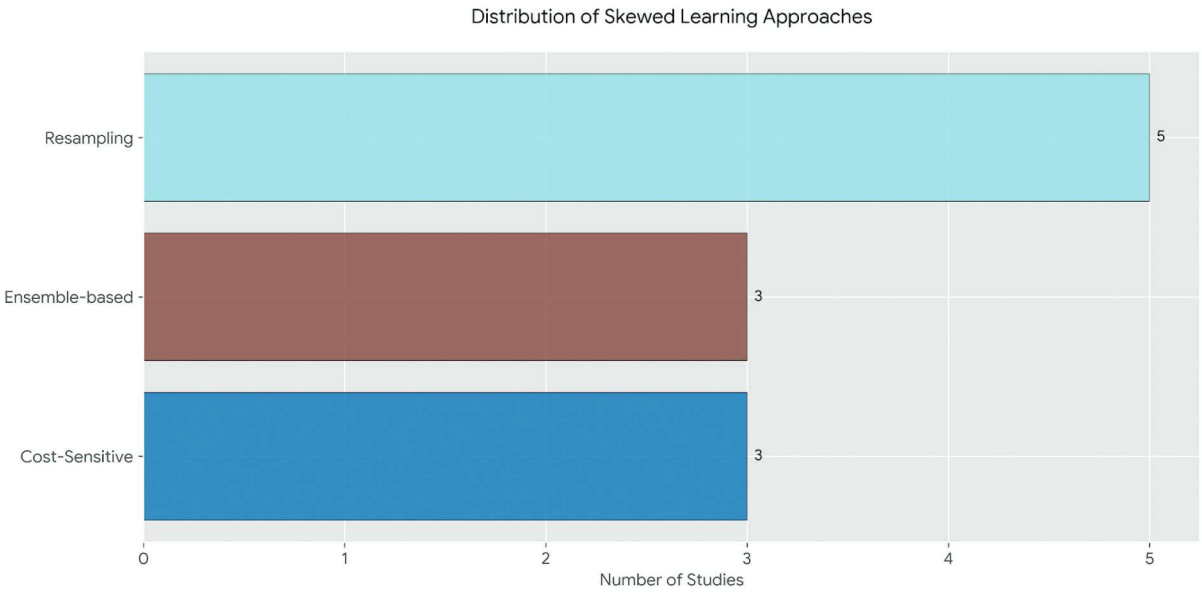

Figure 10. Distribution of skewed learning approaches.

Suitability in real-world applications

EML techniques have demonstrated applicability across a broad spectrum of domains. In image classification, evolutionary algorithms evolve feature extractors and neural architectures for higher efficiency and accuracy. In time-series prediction and anomaly detection, evolutionary strategies optimize parameters and

ensemble structures, enhancing robustness to non-stationary data. Applications also extend to medical diagnosis, spam detection, cybersecurity, and financial forecasting, all of which require adaptability under uncertainty.

Concept drift appears as the most represented domain across the studies, while other fields such as image classification, medical diagnosis, and case-based reasoning (CBR) are found. This diversity highlights the flexibility of EML in addressing visual recognition, sequential modeling, and dynamic decision-making problems.

Emerging application areas include activity recognition, cybersecurity, credit scoring, spam detection, and predictive maintenance, where EML improves adaptability and model generalization. In particular, in cybersecurity, evolutionary methods improve resistance to adversarial attacks, while in medical domains, they optimize diagnostic models for better sensitivity and specificity. Less conventional but promising applications include behavior analysis, microprocessor inspection, Othello gameplay, and gradient-based optimization, emphasizing EML's potential for innovation in complex or evolving environments.

Several studies also explore socially significant use cases such as crime prediction, malaria vector classification, and attack detection, underscoring EML's potential in public safety and epidemiology. These problems often involve heterogeneous data sources (e.g., sensors, text, and social networks), where evolutionary models effectively integrate multimodal signals. Furthermore, the use of abstract problem categories (e.g., CBR, online learning, regression, XOR problems) illustrates EML's expanding role in advancing learning paradigms rather than addressing isolated tasks.

Finally, computational efficiency varies considerably among EML methods. Lightweight approaches such as LASSO and OWA (O(n) complexity) are well-suited for real-time or edge applications, whereas ensemble and dynamic expansion methods (e.g., SEOA, DeDNN) are computationally intensive, requiring high-performance infrastructure. Balancing accuracy and latency is particularly critical for time-sensitive applications such as autonomous systems, where delays under 100 ms are often required.

Illustrative benchmark-based applications

The reviewed literature demonstrates the applicability of EML techniques across a range of dynamic environments where data distributions evolve over time. Rather than focusing on isolated challenges, these applications illustrate how adaptive learning mechanisms can maintain model performance under changing conditions.

Data drift in predictive maintenance

A representative example of data drift can be found in industrial predictive maintenance systems. Adaptive-SPLL (Klein and Verbeke 2020) was evaluated using the NASA C-MAPSS dataset, which contains sensor measurements collected from aircraft engines operating under varying conditions. As engines age and operating environments change, the statistical properties of sensor readings evolve over time, resulting in data drift. Adaptive-SPLL addresses this challenge through adaptive windowing and density estimation techniques that continuously monitor changes in the input distribution. The study demonstrated that adaptive drift detection can identify distributional shifts while maintaining efficient monitoring performance, illustrating the importance of data-drift-aware learning mechanisms in long-term industrial deployments.

Concept drift in streaming data

Several concept-drift methods have been evaluated in synthetic and real-world data streams, where an example, SEOA (Guo, Zhang, and Wang 2021) addresses the concept-drift challenge through a self-evolving online ensemble that dynamically updates classifier composition and weighting as new data become available. Evaluation across multiple benchmark streams exhibiting abrupt, gradual, and recurring drift patterns (SEA, Hyperplane, RBF, LED, Tree, Electricity, KDDCup99, CoverType, and Weather) demonstrated the effectiveness of adaptive ensemble management in preserving predictive performance under changing data distributions. From an EML perspective, SEOA exemplifies how continuous adaptation

mechanisms enable learning systems to remain operational without requiring complete retraining when environmental conditions evolve.

Catastrophic forgetting in class-incremental image classification

A representative example of catastrophic forgetting can be observed in class-incremental image classification. In the iCaRL framework (Rebuffi et al. 2017), a model is trained sequentially as new image classes become available over time. When the network is updated using only data from newly introduced classes, its performance on previously learned classes deteriorates significantly because the parameter updates overwrite earlier representations. To mitigate this effect, iCaRL stores a small set of exemplar images from previously learned classes and combines them with new training data while applying knowledge distillation to preserve earlier predictions. This approach demonstrates the central challenge of catastrophic forgetting in EML: balancing the acquisition of new knowledge with the retention of previously learned information under continuous adaptation.

Online learning for imbalanced data streams

Methods for online learning are increasingly important in applications where data arrive continuously and class distributions are highly imbalanced. Wen Wang, Guo, and Cheng Yan Y (2021) proposed an Online Weighted Sequential Extreme Learning Machine (OWSELM), which extends the Online Sequential Extreme Learning Machine (OS-ELM) by incorporating class-dependent weighting to improve the recognition of minority-class instances. The method was evaluated on several imbalanced benchmark datasets, including Pima, Yeast, and Haberman, among others. Results demonstrated improved classification performance on minority classes while maintaining the computational efficiency required for real-time data stream processing. These findings highlight the importance of cost-sensitive learning mechanisms for addressing class imbalance in evolving data environments.

Adaptive learning in non-stationary environments

Adaptive detection and learning mechanisms are particularly important in domains where data characteristics evolve over time. DetectA (Escovedo et al. 2018) was developed to identify and respond to concept drift in streaming data and was evaluated on benchmark datasets including SEA Concepts, Nebraska, Electricity, CoverType, and Poker Hand. These datasets encompass a variety of drift scenarios, allowing the method to be assessed under changing operating conditions. The reported results demonstrated the ability of DetectA to identify distributional changes and adapt the learning process accordingly, helping to maintain predictive performance as data characteristics evolved. From an EML perspective, this work illustrates the importance of drift-aware adaptation mechanisms for sustaining model effectiveness in dynamic environments.

Table 5 summarizes representative applications drawn from the reviewed literature. Rather than providing an exhaustive catalog, these examples illustrate how the four core EML challenges manifest in different domains and how adaptive learning mechanisms have been employed to address them. The selected studies highlight the diversity of application contexts, ranging from predictive maintenance and streaming analytics to class-incremental learning and imbalanced data streams. However, the literature remains fragmented, with most studies focusing on individual challenges rather than integrated solutions that simultaneously address data drift, concept drift, catastrophic forgetting, and skewed learning. This observation reinforces the need for the unified perspective on EML as proposed in this survey.

Table 5. Representative applications of EML techniques across the four core challenges.

Challenge	Method	Domain	EML Mechanism
Data Drift	Adaptive-SPLL	Predictive Maintenance	Adaptive windowing
Concept Drift	SEOA	Streaming Analytics	Self-evolving ensembles
Concept Drift	DetectA	Stream Learning	Proactive drift detection
Catastrophic Forgetting	iCaRL	Incremental Learning	Replay + distillation
Skewed Learning	OWSELM	Imbalanced Streams	Cost-sensitive weighting

Future research directions

EML is still at an early stage, and several promising avenues require sustained investigation. Our survey identifies a set of future directions that build directly on the limitations of current methods and the gaps highlighted in the comparative analysis.

Dedicated data drift adaptation

While concept drift has received considerable attention, comparatively fewer studies address data drift as a primary challenge. Future research should investigate methods capable of distinguishing between covariate shifts, transient fluctuations, and persistent distributional changes, while providing robust adaptation mechanisms that operate independently of concept-drift assumptions.

Explainability in adaptive systems

While explainable AI is an established research field, little work addresses how explanations should evolve alongside the model itself. Current approaches to explainability assume static models, which limits their applicability in dynamic contexts where decision boundaries change over time. Future research must explore temporal explainability, allowing practitioners to track how predictions and model structures evolve under data drift or catastrophic forgetting. Such methods would not only improve trust but also support regulatory compliance, as the EU AI Act emphasizes traceability in adaptive systems. A key challenge is balancing the trade-off between interpretability and adaptation speed in real-time applications.

Multimodal and cross-domain EML

Most evolving systems to date have been developed for unimodal data streams, often tabular or image-based. Yet many critical applications such as autonomous driving, healthcare monitoring, or industrial IoT generate multimodal streams where visual, textual, and sensor inputs evolve at different rates. Research is needed to develop adaptive fusion strategies that align drifts across modalities without destabilizing the overall model. Cross-domain adaptability also remains underexplored: models that can evolve simultaneously across related but distinct domains (e.g., adapting fraud detection models trained in one country to another with minimal retraining) would have significant practical value.

Benchmarks and evaluation protocols

Our comparative analysis highlights inconsistent evaluation practices: studies use different datasets, metrics, and drift scenarios, limiting comparability. Establishing standardized evolving benchmarks covering synthetic, real-world, and multimodal streams would enable fairer comparisons. Beyond accuracy, benchmarks should evaluate adaptation delay, resource efficiency, fairness, and robustness. Platforms like OpenML could host evolving datasets with community-agreed protocols, ensuring reproducibility. Without such standards, progress risks becoming fragmented and difficult to measure.

Efficiency and resource awareness

Many EML algorithms, especially deep evolving models, are computationally expensive, hindering deployment on edge devices such as wearables or mobile robots. Future work should focus on resource-aware EML, explicitly optimizing latency, memory, and energy consumption alongside accuracy. Promising directions include lightweight drift detectors, sparsity-inducing regularizers, and neuromorphic or FPGA-friendly architectures. Efficiency will be especially crucial in real-time safety-critical environments like autonomous vehicles or medical monitoring, where high latency may undermine system reliability.

Human-in-the-loop EML

Fully autonomous adaptation may not always be acceptable, particularly in safety-critical domains. Future work should, therefore, investigate human-in-the-loop frameworks where domain experts supervise, validate, or guide adaptation. Such approaches could mitigate risks of erroneous drift detection or biased updates. A research challenge is designing interaction protocols that allow human oversight without creating excessive cognitive burden or delaying adaptation in real-time applications.

Ethical and societal dimensions

As EML systems become more pervasive, ethical issues move to the forefront. Adaptation itself may introduce or amplify bias if minority groups are underrepresented in evolving data streams. Future research should embed fairness-aware mechanisms within adaptation pipelines, for example through bias-sensitive drift detectors or regularizers. Transparency will also be essential, as evolving systems must remain auditable even as they change over time. Beyond technical solutions, collaboration with policymakers will be required to align EML with emerging governance frameworks.

Taken together, these directions indicate that the next wave of EML research should not only pursue algorithmic advances, but also tackle system-level integration balancing adaptability, interpretability, efficiency, fairness, and human oversight. Only by addressing these intertwined challenges can EML transition from an emerging research niche into a trusted paradigm for dynamic, real-world AI.

Conclusion

This survey analyzed extensively influential works on EML methods and techniques, specifically targeting critical challenges: data drift, concept drift, catastrophic forgetting, and skewed data. The extensive coverage includes 26 detailed surveys spanning all challenges and providing a foundational perspective on existing methodologies. Data drift, addressed through seven key studies, highlighted innovative drift detection methods, adaptive data augmentation, and active learning approaches crucial for maintaining performance consistency in dynamically evolving data environments. The extensive exploration of concept drift with 61 studies emphasized effective strategies such as sliding windows for timely drift detection, ensemble-based adaptations for robust predictions, meta-learning methods facilitating rapid adaptation, and prototype-based learning approaches to manage evolving data distributions effectively.

The critical issue of catastrophic forgetting was carefully examined across 34 significant studies, detailing approaches like dynamic network expansion strategies promoting adaptable model architectures, and Elastic Weight Consolidation to promote a fairer distribution of weights for old values, helping them to maintain a grade of importance over the time. Meanwhile, skewed data challenges, discussed through 14 dedicated papers, underlined the efficacy of resampling techniques, cost-sensitive learning, ensemble methods, and fairness-aware models to address class imbalance and fairness constraints effectively.

Evaluation methodologies across these studies were diverse, incorporating a broad spectrum of synthetic and real-world datasets. Synthetic datasets allowed controlled simulation of specific drift conditions, catastrophic forgetting scenarios, and skewed distributions, enabling rigorous benchmarking of algorithms. Real-world datasets further validated the practical applicability of these methods in complex, uncontrolled environments such as healthcare, finance, cybersecurity, and natural language processing. Common evaluation metrics included accuracy, precision, recall, F1-score, area under the ROC curve (AUC-ROC), and fairness metrics, which collectively provided comprehensive performance assessments under various evolving scenarios.

From a scientific impact perspective, this survey reveals the growing influence of EML research across both foundational and applied domains. Advances in drift detection and adaptive learning have informed the design of autonomous systems capable of long-term operation without retraining, while continual learning and fairness-aware frameworks have contributed to more responsible and explainable AI. The surveyed studies demonstrate strong cross-domain relevance, driving innovation in industrial process monitoring, medical diagnostics, predictive maintenance, and adaptive decision-making systems. These contributions reinforce the transformative potential of EML not only for improving algorithmic robustness but also for enabling sustainable, adaptive, and ethically aligned intelligent systems.

Overall, this comprehensive survey demonstrates significant advancements and remaining challenges in EML methods, highlighting future research opportunities particularly in integrating multiple adaptive strategies, enhancing real-time applicability, and ensuring fairness and robustness in practical, real-world applications.

Acknowledgements

For the purpose of open access, the authors have applied a Creative Commons Attribution (CC BY) licence to any Author Accepted Manuscript version arising from this submission.

Author contributions

CRediT: **Ignacio Cabrera Martin:** Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft; **Subhaditya Mukherjee:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – review & editing; **Almas Baimagambetov:** Investigation, Validation, Writing – review & editing; **Joaquin Vanschoren:** Conceptualization, Funding acquisition, Project administration, Resources, Validation, Writing – review & editing; **Nikolaos Polatidis:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

Funded by the European Union Horizon Europe programme through ALFIE under Grant Agreement [101177912.] Views and opinions expressed are, however, those of the author(s) only and do not necessarily reflect those of the European Union or the Agency. Neither the European Union nor the granting authority can be held responsible for them.

ORCID

Nikolaos Polatidis http://orcid.org/0000-0003-4249-4953

Data availability statement

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Ethics approval

Not applicable. This study does not involve human participants, animals, or any personal or sensitive data.

References

Agrawal, A., M. Sazos, A. A. Durra, et al. 2020. "Towards Robust Power Grid Attack Protection Using Lightgbm with Concept Drift Detection and Retraining." Proceedings of the 2020 Joint Workshop on CPS&IoT Security and Privacy (CPSIOTSEC'20), New York, NY, USA: 31–36. ACM. https://doi.org/10.1145/3411498.3419964.

Alcalá-Fdez, J., A. Fernández, J. Luengo, et al. 2011. "Keel Data-Mining Software Tool: Data Set Repository, Integration of Algorithms and Experimental Analysis Framework." *Journal of Information Sciences* 17:255–287. https://www.oldcitypublishing.com/journals/mvlsc-home/mvlsc-issue-contents/mvlsc-volume-17-number-2-3-2011/mvlsc-17-2-3-p-255-287/.

Aleixo, E. L., J. G. Colonna, M. Cristo, and E. Fernandes. 2024. "Catastrophic Forgetting in Deep Learning: A Comprehensive Taxonomy." *Journal of the Brazilian Computer Society* 30 (1): 175–211. https://doi.org/10.5753/jbcs.2024.3966.

AlQabbany, A. O., and A. M. Azmi. 2021. "Measuring the Effectiveness of Adaptive Random Forest for Handling Concept Drift in Big Data Streams." *Entropy* 23 (7): 859. https://doi.org/10.3390/e23070859.

Aminian, E., R. P. Ribeiro, and J. Gama. 2021. "Chebyshev Approaches for Imbalanced Data Streams Regression Models." *Data Mining and Knowledge Discovery* 35 (6): 2389–2466. https://doi.org/10.1007/s10618-021-00793-1.

Anderson, R., Y. S. Koh, and G. Dobbie. 2018. "Lift-Per-Drift: An Evaluation Metric for Classification Frameworks with Concept Drift Detection." In *AI 2018: Advances in Artificial Intelligence*, edited by T. Mitrovic, B. Xue, and X. Li, 630–642. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-03991-257.

Angbera, A., and H. Y. Chan. 2024. "Sabedm: A Sliding Adaptive Beta Distribution Model for Concept Drift Detection in a Dynamic Environment." *Knowledge and Information Systems* 66 (3): 2039–2062. https://doi.org/10.1007/s10115-023-02004-3.

Angelov, P. 2023. "Evolving Fuzzy Systems." In *Encyclopedia of Machine Learning and Data Mining*, edited by C. and G. I. Webb, 725–741. New York, NY: Springer. https://doi.org/10.1007/978-1-0716-2628-3192.

Angelov, P., D. P. Filev, and N. Kasabov. 2010. *Evolving Intelligent Systems: Methodology and Applications*. New York, NY, USA: IEEE Press Series on Computational Intelligence, John Wiley & Sons. https://doi.org/10.1002/9780470569962.

Angelov, P., and X. Zhou. 2006. "Evolving Fuzzy Systems from Data Streams in Real-Time." Proceedings of the 2006 International Symposium on Evolving Fuzzy Systems, Lake District, UK: 29–35. Ambleside: IEEE. https://eprints.lancs.ac.uk/id/eprint/938/1/AngelovZhou144.pdf.

Ashfahani, A., and M. Pratama. 2019. "Autonomous Deep Learning: Continual Learning Approach for Dynamic Environments." In *Proceedings of the 2019 SIAM International Conference on Data Mining (SDM)*, 666–674. Calgary, Alberta, Canada: SIAM.

Ashok, S., S. Ezhumalai, and T. Patwa. 2023. "Remediating Data Drifts and Re-Establishing Ml Models." *Procedia Computer Science* 218:799–809. 01. https://doi.org/10.1016/j.procs.2023.01.060.

Baidari, I., and N. Honnikoll. 2021. "Bhattacharyya Distance Based Concept Drift Detection Method for Evolving Data Stream." *Expert Systems with Applications* 183:115303. https://doi.org/10.1016/j.eswa.2021.115303.

Baier, L., T. Schlö, J. Schöffer, et al. 2022. "Detecting Concept Drift with Neural Network Model Uncertainty." arXiv preprint arXiv:2107.01873. https://doi.org/10.48550/arXiv.2107.01873.

Balestra, C., B. Li, and E. Müller. 2023. "Slidshaps – Sliding Shapley Values for Correlation-Based Change Detection in Time Series." In *2023 IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA)*, 1–10. Thessaloniki, Greece. https://doi.org/10.1109/DSAA60987.2023.1030263.

Barros, R. S. M., and S. G. T. C. Santos. 2018. "A Large-Scale Comparison of Concept Drift Detectors." *Information Sciences* 451-452:348–370. https://doi.org/10.1016/j.ins.2018.04.014.

Bayram, F., B. S. Ahmed, and A. Kassler. 2022. "From Concept Drift to Model Degradation: An Overview on Performance-Aware Drift Detectors." *Knowledge-Based Systems* 245:108632. arXiv preprint arXiv:2203.11070. https://doi.org/10.1016/j.knosys.2022.108632.

Bhatnagar, A., and D. Gajjar. 2025. "Policy Implications of Artificial Intelligence (Ai)." Accessed March 5, 2025. https://post.parliament.uk/research-briefings/post-pn-0708/UKParliamentResearchBriefing.

Blackard, Jock. 1998. *Covertype*. CI Machine Learning Repository. https://doi.org/10.24432/C50K5N.

Bu, L., C. Alippi, and D. Zhao. 2018. "A Pdf-Free Change Detection Test Based on Density Difference Estimation." *IEEE Transactions on Neural Networks and Learning Systems* 29 (2): 324–334. https://doi.org/10.1109/TNNLS.2016.2619909.

Buzzega, P., M. Boschini, A. Porrello, et al. 2021. "Rethinking Experience Replay: A Bag of Tricks for Continual Learning." In *Proceedings of the 25th International Conference on Pattern Recognition (ICPR)*, 2180–2187. Los Alamitos, CA, USA. https://doi.org/10.1109/ICPR48806.2021.9412614.

Cabrera Martin, I., S. Mukherjee, A. Baimagambetov, et al. 2025. "Evolving Machine Learning: A Unified Survey of Drift, Forgetting, and Adaptation. arXiv preprint arXiv:2505.17902 https://arxiv.org/abs/2505.17902.

Carta, A., A. Cossu, F. Errica, and D. Bacciu. 2022. "Catastrophic Forgetting in Deep Graph Networks: A Graph Classification Benchmark." *Frontiers in Artificial Intelligence* 5:824655. https://doi.org/10.3389/frai.2022.824655.

Cassales, G., H. Gomes, A. Bifet, B. Pfahringer, and H. Senger. 2021. "Improving the Performance of Bagging Ensembles for Data Streams Through Mini-Batching." *Information Sciences* 580:260–282. https://doi.org/10.1016/j.ins.2021.08.085.

Castellani, A., S. Schmitt, and B. Hammer. 2021. "Task-Sensitive Concept Drift Detector with Constraint Embedding." In *2021 IEEE Symposium Series on Computational Intelligence (SSCI)*, 1–8. Orlando, FL, USA. https://doi.org/10.1109/SSCI50451.2021.9659969.

Cattral, R., and F. Oppacher. 2002. "Poker Hand." (UCI) Machine Learning Repository. https://doi.org/10.24432/C5KW38.

Cavenaghi, E., G. Sottocornola, F. Stella, and M. Zanker. 2021. "Non Stationary Multi-Armed Bandit: Empirical Evaluation of a New Concept Drift-Aware Algorithm." *Entropy* 23 (3): 380. https://doi.org/10.3390/e23030380.

Celik, B., and J. Vanschoren. 2021. "Adaptation Strategies for Automated Machine Learning on Evolving Data." *IEEE Transactions on Pattern Analysis & Machine Intelligence* 43 (9): 3067–3078. https://doi.org/10.1109/TPAMI.2021.3062900.

Chambers, L., M. Gaber, and H. Ghomeshi. 2025. "Deepstreamensemble: Streaming Adaptation to Concept Drift in Deep Neural Networks." *International Journal of Machine Learning and Cybernetics* 16 (5–6): 3955–3976. https://doi.org/10.1007/s13042-024-02492-x.

Chapelin, J., A. Voisin, B. Rose, B. Iung, L. Steck, L. Chaves, M. Lauer, and O. Jotz. 2025. "Data-Driven Drift Detection and Diagnosis Framework for Predictive Maintenance of Heterogeneous Production Processes: Application to

a Multiple Tapping Process." *Engineering Applications of Artificial Intelligence* 139:109552. https://doi.org/10.1016/j.engappai.2024.109552.
Chen, Z., M. Han, H. Wu, M. Li, and X. Zhang. 2023. "A Multi-Level Weighted Concept Drift Detection Method." *Journal of Supercomputing* 79 (5): 5154–5180. https://doi.org/10.1007/s11227-022-04864-y.
Chen, Z., V. Sheng, A. Edwards, and K. Zhang. 2023. "An Effective Cost-Sensitive Sparse Online Learning Framework for Imbalanced Streaming Data Classification and Its Application to Online Anomaly Detection." *Knowledge and Information Systems* 65 (1): 59–87. https://doi.org/10.1007/s10115-022-01745-x.
Cortez ACP. 2009. "Wine Quality." UCI Machine Learning Repository. https://doi.org/10.24432/C56S3T.
Costa, A., R. Giusti, dos Santos E. M., and E. M. dos Santos. 2025. "Analysis of Descriptors of Concept Drift and Their Impacts." *Informatics* 12 (1): 13. https://doi.org/10.3390/informatics12010013.
Czarnowski, I. 2021. "Learning from Imbalanced Data Streams Based on Over-Sampling and Instance Selection." In *Computational Science – ICCS 2021*, edited by M. Paszynski, D. Kranzlmüller, V. V. Krzhizhanovskaya, J. J. Dongarra, and P. M. A. Sloot, 378–391. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-77967-2. 32.
Czarnowski, I. 2022. "Weighted Ensemble with One-Class Classification and Over-Sampling and Instance Selection (Wecoi): An Approach for Learning from Imbalanced Data Streams." *Journal of Computational Science* 61:101614. https://doi.org/10.1016/j.jocs.2022.101614.
de Barros R. S. M., de Carvalho Santos S. G. T., R. S. M. D. Barros, and S. G. T. D. C. Santos. 2019. "An Overview and Comprehensive Comparison of Ensembles for Concept Drift." *Information Fusion* 52:213–244. https://doi.org/10.1016/j.inffus.2019.03.006.
De Lange, M., R. Aljundi, M. Masana, S. Parisot, X. Jia, A. Leonardis, G. Slabaugh, and T. Tuytelaars. 2022. "A Continual Learning Survey: Defying Forgetting in Classification Tasks." *IEEE Transactions on Pattern Analysis & Machine Intelligence* 44 (7): 3366–3385. https://doi.org/10.1109/TPAMI.2021.3057446.
de Lima Cabral D. R., and de Barros R. S. M. 2018. "Concept Drift Detection Based on fisher's Exact Test." *Information Sciences*: 220–234. https://doi.org/10.1016/j.ins.2018.02.054.
de Mello R. F., Y. Vaz, C. H. Grossi, R. F. de Mello, and A. Bifet. 2019. "On Learning Guarantees to Unsupervised Concept Drift Detection on Data Streams." *Expert Systems with Applications* 117:90–102. https://doi.org/10.1016/j.eswa.2018.08.054.
Deng, L. 2012. "The Mnist Database of Handwritten Digit Images for Machine Learning Research [Best of the Web]." *IEEE Signal Processing Magazine* 29 (6): 141–142. https://doi.org/10.1109/MSP.2012.2211477.
Dosovitskiy, A., L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, and M. Dehghani. 2021. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." CoRR abs/2010.11929, arXiv:2010.11929. https://doi.org/10.48550/arXiv.2010.
Elkhawaga, G., M. Abuelkheir, S. I. Barakat, A. M. Riad, and M. Reichert. 2020. "Conda-pm—a Systematic Review and Framework for Concept Drift Analysis in Process Mining." *Algorithms* 13 (7): 161. https://doi.org/10.3390/a13070161.
Escovedo, T., A. Koshiyama, A. A. da Cruz, and M. Vellasco. 2018. "Detecta: Abrupt Concept Drift Detection in Non-Stationary Environments." *Applied Soft Computing* 62:119–133. https://doi.org/10.1016/j.asoc.2017.10.031.
European Parliament and Council of the European Union. 2024. "Regulation (Eu) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)." Accessed March 5, 2025. http://data.europa.eu/eli/reg/2024/1689/oj/eng.
Fan, Q., C. Liu, Y. Zhao, and Y. Li. 2023. "Unsupervised Online Concept Drift Detection Based on Divergence and Ewma." In *Web and Big Data*, B. Li, L. Yue, C. Tao, X. Han, D. Calvanese, and T. Amagasa, 121–134. Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-25158-0.
Fan, W., and S. Stolfo. 1999. "KDD Cup 1999 Data." UCI Machine Learning Repository. https://doi.org/10.24432/C51C7N.
Folino, G., M. Guarascio, and G. Papuzzo. 2019. "Exploiting Fractal Dimension and a Distributed Evolutionary Approach to Classify Data Streams with Concept Drifts." *Applied Soft Computing* 75:284–297. https://doi.org/10.1016/j.asoc.2018.11.009.
Friedrich, B., T. Sawabe, and A. Hein. 2023. "Unsupervised Statistical Concept Drift Detection for Behaviour Abnormality Detection." *Applied Intelligence* 53 (3): 2527–2537. https://doi.org/10.1007/s10489-022-03611-3.
Gemaque, R. N., A. F. J. Costa, R. Giusti, and E. M. dos Santos. 2020. "An Overview of Unsupervised Drift Detection Methods." *WIREs Data Mining and Knowledge Discovery* 10 (6): e1381. https://doi.org/10.1002/widm.1381.
Gijsbers, P., M. L. P. Bueno, and S. Coors. 2024. "Amlb: An Automl Benchmark." *The Journal of Machine Learning Research* 25 (1): 1–65. https://www.jmlr.org/papers/v25/22-0493.html.
Gözüac¸ık, Ö., A. Büyük¸cakır, and H. Bonab. 2019. "Unsupervised Concept Drift Detection with a Discriminative Classifier." Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM '19), New York, NY, USA: 2365–2368. ACM. https://doi.org/10.1145/3357384.3358144.
Gözüaçık, Ö., and F. Can. 2021. "Concept Learning Using One-Class Classifiers for Implicit Drift Detection in Evolving Data Streams." *Artificial Intelligence Review* 54 (5): 3725–3747. https://doi.org/10.1007/s10462-020-09939-x.
Guo, H., H. Li, Q. Ren, and W. Wang. 2022. "Concept Drift Type Identification Based on Multi-Sliding Windows." *Information Sciences* 585:1–23. https://doi.org/10.1016/j.ins.2021.11.023.

Guo, H., Z. Wu, Q. Ren, and W. Wang. 2025. "Accelerating the Convergence of Concept Drift Based on Knowledge Transfer." *Pattern Recognition* 159:111145. https://doi.org/10.1016/j.patcog.2024.111145.
Guo, H., H. Xia, H. Li, and W. Wang. 2023. "Concept Evolution Detection Based on Noise Reduction Soft Boundary." *Information Sciences* 628:391–408. https://doi.org/10.1016/j.ins.2023.01.115. ins.
Guo, H., S. Zhang, and W. Wang. 2021. "Selective Ensemble-Based Online Adaptive Deep Neural Networks for Streaming Data with Concept Drift." *Neural Networks* 142:437–456. https://doi.org/10.1016/j.neunet.2021.06.027.
Han, M., F. Meng, and C. Li. 2024. "Variance Feedback Drift Detection Method for Evolving Data Streams Mining." *Applied Sciences* 14 (16): 7157. https://doi.org/10.3390/app14167157.
High-Level Expert Group on Artificial Intelligence. 2019. "Ethics Guidelines for Trust- Worthy Ai — Shaping europe's Digital Future." March 6. https://digitalstrategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai.
Hinder, F., V. Vaquet, and B. Hammer. 2022. "Suitability of Different Metric Choices for Concept Drift Detection." In *Advances in Intelligent Data Analysis XX*, edited by T. Bouadi, E. Fromont, and E. Hüllermeier, 157–170. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-031-01333-1.
Hinder, F., V. Vaquet, and B. Hammer. 2023. "One or Two Things We Know About Concept Drift — a Survey on Monitoring Evolving Environments." *Frontiers in Artificial Intelligence* 7. https://doi.org/10.3389/frai.2024.1330257. arXiv.2310.15826, arXiv preprint arXiv:2310.15826.
Hovakimyan, G., and J. M. Bravo. 2024. "Evolving Strategies in Machine Learning: A Systematic Review of Concept Drift Detection." *Information* 15 (12): 786. https://doi.org/10.3390/info15120786.
Iman, M., J. A. Miller, and K. Rasheed. 2023. "Expanse: A Deep Continual / Progressive Learning System for Deep Transfer Learning." *Technologies* 11 (2): 40. https://doi.org/10.3390/technologies11020040.
Iwashita, A. S., de Albuquerque V. H. C., J. P. Papa, and V. H. C. de Albuquerque. 2019. "Learning Concept Drift with Ensembles of Optimum-Path Forest-Based Classifiers." *Future Generation Computer Systems* 95:198–211. https://doi.org/10.1016/j.future.2019.01.005.
Iwashita, A. S., and J. P. Papa. 2019. "An Overview on Concept Drift Learning." *IEEE Access* 7:1532–1547. https://doi.org/10.1109/ACCESS.2018.2886026.
Jaworski, M., L. Rutkowski, and P. Angelov. 2020. "Concept Drift Detection Using Autoen-Coders in Data Streams Processing." In *Artificial Intelligence and Soft Computing*, edited by L. Rutkowski, R. Scherer, R. Tadeusiewicz, L. A. Zadeh, and J. M. Zurada, 124–133. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-61401-012.
Jiang, M., J. Fan, and F. Li. 2025. "Advances in Continual Learning: A Comprehensive Review." *Expert Systems with Applications* 294:128739. https://doi.org/10.1016/j.eswa.2025.128739.
Johnson, J. M., and T. M. Khoshgoftaar. 2019. "Survey on Deep Learning with Class Imbalance." *Journal of Big Data* 6 (1): 27. https://doi.org/10.1186/s40537-019-0192-5.
Kaminskyi, D., B. Li, and E. Müller. 2022. "Reconstruction-Based Unsupervised Drift Detection Over Multivariate Streaming Data." In *2022 IEEE International Conference on Data Mining Workshops (ICDMW)*, 807–813. Orlando, FL, USA. https://doi.org/10.1109/ICDMW58026.2022.00109.
Kasabov, N. 2007. *Evolving Connectionist Systems: The Knowledge Engineering Approach*. 2nd ed. London, UK: Springer. https://doi.org/10.1007/978-1-84628-347-5.
Kelly, M., R. Longjohn, and K. Nottingham. 2023. "The Uci Machine Learning Repository." https://archive.ics.uci.edu.
Kim, B., and B. Jeon. 2025. "Exploring Feasibility of Data Drift Detection via In-Stream Data for Vision Models." In *2025 19th International Conference on Ubiquitous Information Management and Communication (IMCOM)*, 1–2. Bangkok, Thailand. https://doi.org/10.1109/IMCOM64595.2025.10857506.
Kirkpatrick, J., R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, et al. 2017. "Overcoming Catastrophic Forgetting in Neural Networks." *Proceedings of the National Academy of Sciences of the United States of America* 114 (13): 3521–3526. https://doi.org/10.1073/pnas.1611835114.
Klein, S., and M. Verbeke. 2020. "An Unsupervised Methodology for Online Drift Detection in Multivariate Industrial Datasets." In *2020 International Conference on Data Mining Workshops (ICDMW)*, 392–399. Sorrento, Italy. https://doi.org/10.1109/ICDMW51313.2020.0006.
Komorniczak, J., P. Zyblewski, and P. Ksieniewicz. 2022. "Statistical Drift Detection Ensemble for Batch Processing of Data Streams." *Knowledge-Based Systems* 252:109380. https://doi.org/10.1016/j.knosys.2022.109380.
Korycki, L., and B. Krawczyk. 2019. "Unsupervised drift detector ensembles for data stream mining." In *2019 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 317–325. Washington, DC, USA. https://doi.org/10.1109/DSAA.2019.00047.
Krawczyk, B., B. Pfahringer, and M. Woźniak. 2018. "Combining Active Learning with Concept Drift Detection for Data Stream Mining." In *2018 IEEE International Conference on Big Data (Big Data)*, 2239–2244. Seattle, WA, USA. https://doi.org/10.1109/BigData.2018.8622549.
Krizhevsky, A. 2009. Learning Multiple Layers of Features from Tiny Images. Tech. Rep. University of Toronto.
Kumar, P., R. Bhatnagar, and K. Gaur, 2021. "Classification of Imbalanced Data: Review of Methods and Applications." *IOP Conference Series: Materials Science and Engineering*: 1099 (1): 012077. https://doi.org/10.1088/1757-899X/1099/1/012077.
Lee, S. W., J. H. Kim, and J. Jun. 2017. "Overcoming Catastrophic Forgetting by Incremental Moment Matching." *Advances in Neural Information Processing Systems (NeurIPS)*. https://doi.org/10.48550/arXiv.1703.08475.

Li, B., Y. Wang, and W. Liu. 2025. "Towards Understanding Catastrophic Forgetting in Two-Layer Convolutional Neural Networks." *OpenReview*. https://openreview.net/forum?id=WkqZ6Qmmq2. provides a theoretical analysis of catastrophic forgetting in two-layer convolutional neural networks under continual learning.

Li, J., S. Fong, R. K. Wong, and V. W. Chu. 2018. "Adaptive Multi-Objective Swarm Fusion for Imbalanced Data Classification." *Information Fusion* 39:1–24. 03. https://doi.org/10.1016/j.inffus.2017.03.007.

Li, J., H. Yu, Z. Zhang, X. Luo, and S. Xie. 2024. "Concept Drift Adaptation by Exploiting Drift Type." *ACM Transactions on Knowledge Discovery from Data* 18 (4): 22. https://doi.org/10.1145/3638777.

Li, Q., Q. Xiong, S. Ji, Y. Yu, C. Wu, and M. Gao. 2021. "Incremental Semi-Supervised Extreme Learning Machine for Mixed Data Stream Classification." *Expert Systems with Applications* 185:115591. https://doi.org/10.1016/j.eswa.2021.115591.

Liu, A., J. Lu, F. Liu, and G. Zhang. 2018. "Accumulating Regional Density Dissimilarity for Concept Drift Detection in Data Streams." *Pattern Recognition* 76:256–272. https://doi.org/10.1016/j.patcog.2017.11.009.

Logacjov, A., K. Bach, A. Kongsvold, H. B. Bårdstu, and P. J. Mork. 2021. "Harth: A Human Activity Recognition Dataset for Machine Learning." *Sensors* 21 (23): 7853. https://doi.org/10.3390/s21237853.

Lopez-Paz, D., and M. Ranzato. 2017. "Gradient Episodic Memory for Continual Learning." *Advances in Neural Information Processing Systems (NeurIPS) 2017*. https://proceedings.neurips.cc/paper/2017/hash/21be9a4c0ee4c59d4f1f699f51c929db-Abstract.html.

Lu, J., A. Liu, F. Dong, et al. 2019. "Learning Under Concept Drift: A Review." *IEEE Transactions on Knowledge and Data Engineering* 31 (12): 2346–2363. https://doi.org/10.1109/TKDE.2018.2876857.

Lughofer, E. 2011. "Evolving Fuzzy Systems: Methodologies, Advanced Concepts and Applications." In *Studies in Fuzziness and Soft Computing*, Vol. 266. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-642-18087-3.

Lukats, D., O. Zielinski, A. Hahn, and F. Stahl. 2025. "A Benchmark and Survey of Fully Unsupervised Concept Drift Detectors on Real-World Data Streams." *International Journal of Data Science and Analytics* 19 (1): 1–31. https://doi.org/10.1007/s41060-024-00620-y.

Mangasarian, O., and W. Wolberg. 1993. "Breast Cancer Wisconsin (Diagnostic)." (UCI) Machine Learning Repository. https://doi.org/10.24432/C5DW2B.

Mardiansyah, H., R. W. Sembiring, and S. Efendi. 2021. "Handling Problems of Credit Data for Imbalanced Classes Using Smotexgboost." In *Journal of Physics: Conference Series*, 012011. Lombok, West Nusa Tenggara, Indonesia. https://doi.org/10.1088/1742-6596/1830/1/012011.

Masana, M., X. Liu, B. Twardowski, M. Menta, A. D. Bagdanov, and J. van de Weijer. 2023. "Class-Incremental Learning: Survey and Performance Evaluation on Image Classification." *IEEE Transactions on Pattern Analysis & Machine Intelligence* 45 (5): 5513–5533. https://doi.org/10.1109/TPAMI.2022.3213473.

Mirzadeh, Iman S., Farajtabar, M., Gorur, D., Pascanu, R. and Ghasemzadeh, H. 2021. Linear Mode Connectivity in Multitask and Continual Learning. In *Proceedings of the Ninth International Conference on Learning Representations (ICLR 2021)*. OpenReview.net. https://openreview.net/forum?id=Fmg_fQYUejf .

Moleda, M., A. Momot, and D. Mrozek. 2020. "Concept Drift and Avoiding Its Negative Effects in Predictive Modeling of Failures of Electricity Production Units in Power Plants." In *2020 28th International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS)*, 1–7. Nice, France. https://doi.org/10.1109/MASCOTS50786.2020.9285972.

Nakai, K. 1991. "Yeast." (UCI) Machine Learning Repository. https://doi.org/10.24432/C5KG68.

Nakai, K. 1996. "Ecoli." (UCI) Machine Learning Repository. https://doi.org/10.24432/C5388M.

Naqvi, N., S. U. Rehman, and M. Z. Islam. 2022. "Windrift: Early Detection of Concept Drift Using Corresponding and Hierarchical Time Windows." In *Data Mining*, edited by L.A.F. Park, H.M. Gomes, M. Doborjeh, Y.L. Boo, Y.S. Koh, Y. Zhao, G. Williams, and S. Simoff, 73–89. Singapore: Springer Nature. https://doi.org/10.1007/978-981-19-8746-5.

Cuong V. Nguyen, Achille, A., Lam, M., Hassner, T., Mahadevan, V. and Soatto, S. 2019. Toward Understanding Catastrophic Forgetting in Continual Learning. In *Meta-Learning Workshop at the 33rd Conference on Neural Information Processing Systems (NeurIPS 2019)*. Vancouver, BC, Canada. Workshop paper; arXiv:1908.01091. https://doi.org/10.48550/arXiv.1908.01091.

Palli, A. S., J. Jaafar, H. M. Gomes, M. A. Hashmani, and A. R. Gilal. 2022. "An Experimental Analysis of Drift Detection Methods on Multi-Class Imbalanced Data Streams." *Applied Sciences* 12 (22): 11688. https://doi.org/10.3390/app122211688.

Parisi, G. I., R. Kemker, J. L. Part, C. Kanan, and S. Wermter. 2019. "Continual Lifelong Learning with Neural Networks: A Review." *Neural Networks* 113:54–71. 01. https://doi.org/10.1016/j.neunet.2019.01.012.

Park, S., and T. Suh. 2022. "Continual Learning with Speculative Backpropagation and Activation History." *IEEE Access* 10:38555–38564. https://doi.org/10.1109/ACCESS.2022.3166158.

Porwik, P., T. Orczyk, K. Wrobel, and B. M. Dadzie. 2025. "A Novel Method for Drift Detection in Streaming Data Based on Measurement of Changes in Feature Ranks." *Journal of Artificial Intelligence and Soft Computing Research* 15 (2): 147–166. https://doi.org/10.2478/jaiscr-2025-0008.

Ramasesh, V. S., A. R. Reibman, and D. D. Cox. 2020. "Anatomy of Catastrophic Forgetting: Hidden Representations and Task Semantics." International Conference on Learning Representations (ICLR) 2020. Virtual Conference. https://openreview.net/forum?idrJeQUnAqYQ,publishedonOpenReview.

Rebuffi, S. A., A. Kolesnikov, and G. Sperl. 2017. "Icarl: Incremental Classifier and Representation Learning." In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2001–2010. Honolulu, HI, USA. https://doi.org/10.1109/CVPR.2017.322cvpr2017/html/RebuffiiCaRLIncrementalClassifierCVPR2017paper.html,introducesexemplar-basedrehearsalforclass-incrementallearning.

Ren, W., T. Zhao, and W. Qin. 2023. "T-Sas: Toward Shift-Aware Dynamic Adaptation for Streaming Data." Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23), New York, NY, USA: 4244–4248. ACM. https://doi.org/10.1145/3583780.3615267.

Riemer, M., I. Cases, and Y. Liu. 2019. "Meta-Experience Replay for Continual Learning." European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD) 2019, Lecture Notes in Computer Science, 554–569. Vol. 11903. New Orleans, LA, USA: Springer. https://doi.org/10.1007/978-3-030-43823-934.

Rusu, A. A., N. C. Rabinowitz, G. Desjardins, et al. 2022. "Progressive Neural Networks." arXiv preprint arXiv:1606.04671. https://doi.org/10.48550/arXiv.1606.04671.

Sakurai, G. Y., J. F. Lopes, B. B. Zarpelão, and S. Barbon Junior. 2023. "Benchmarking Change Detector Algorithms from Different Concept Drift Perspectives." *Future Internet* 15 (5): 169. https://doi.org/10.3390/fi15050169.

Schwartz, R., J. Dodge, N. A. Smith, and O. Etzioni. 2020. "Green ai." *Communications of the ACM* 63 (12): 54–63. https://doi.org/10.1145/3381831.

Senarathna, D., S. Tragoudas, and K. N. Gowda. 2023. "Detection and Quantization of Data Drift in Image Classification Neural Networks." In *2023 IEEE 24th International Conference on High Performance Switching and Routing (HPSR)*, 38–42. Albuquerque, NM, USA. https://doi.org/10.1109/HPSR57248.2023.10147972.

Serra, J., D. Suris, and M. Miron. 2018. "Overcoming Catastrophic Forgetting with Hard Attention to the Task." Proceedings of the 35th International Conference on Machine Learning (ICML). Introduces Hard Attention to the Task (HAT) for mitigating catastrophic forgetting in continual learning, 4548–4557. Stockholm, Sweden. https://proceedings.mlr.press/v80/serra18a.html

Shang, D., G. Zhang, and J. Lu. 2020. "Fast Concept Drift Detection Using Unlabeled Data." *Developments of Artificial Intelligence Technologies in Computation and Robotics World Scientific*: 133–140. https://doi.org/10.1142/9789811223334. 0017.

Shang, D., G. Zhang, and J. Lu. 2025. "Novelty-Aware Concept Drift Detection for Neural Networks." *Neurocomputing* 617:128933. https://doi.org/10.1016/j.neucom.2024.128933.

Shi, H., Z. Xu, H. Wang, W. Qin, W. Wang, Y. Wang, Z. Wang, S. Ebrahimi, and H. Wang. 2025. "Continual Learning of Large Language Models: A Comprehensive Survey." *ACM Computing Surveys* 58 (5): 1–42. https://doi.org/10.1145/3735633.

Siahroudi, S. K., P. Z. Moodi, and H. Beigy. 2018. "Detection of Evolving Concepts in Non- Stationary Data Streams: A Multiple Kernel Learning Approach." *Expert Systems with Applications* 91:187–197. https://doi.org/10.1016/j.eswa.2017.08.033.

Soares, R. G. F., and L. L. Minku. 2025. "Online Ensemble Model Compression for Nonstationary Data Stream Learning." *Neural Networks* 185:107151. https://doi.org/10.1016/j.neunet.2025.107151.

Souza, V. M. A., F. A. Chowdhury, and A. Mueen. 2020. "Unsupervised Drift Detection on High- Speed Data Streams." 2020 IEEE International Conference on Big Data (Big Data), Virtual Conference. 102–111. https://doi.org/10.1109/BigData50022.2020.9377880.

Suárez-Cetrulo, A. L., D. Quintana, and A. Cervantes. 2023. "A Survey on Machine Learning for Recurring Concept Drifting Data Streams." *Expert Systems with Applications* 213:118934. https://doi.org/10.1016/j.eswa.2022.118934.

Sulaiman, M., M. Farmanbar, S. Kagami, A. N. Belbachir, and C. Rong. 2025. "Online Deep learning's Role in Conquering the Challenges of Streaming Data: A Survey." *Knowledge and Information Systems* 67 (4): 3159–3203. https://doi.org/10.1007/s10115-025-02351-3.

Suprem, A., A. Musaev, and C. Pu. 2019. "Concept Drift Adaptive Physical Event Detection for Social Media Streams." In *Services – SERVICES 2019*, edited by Y. Xia and L. J. Zhang, 92–105. Berlin, Heidelberg, Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-23381-5.

Tadros, T., G. P. Krishnan, Ramyaa, R. Ramyaa, and M. Bazhenov. 2022. "Sleep-Like Unsupervised Replay Reduces Catastrophic Forgetting in Artificial Neural Networks." *Nature Communications* 13 (1): 7239. https://doi.org/10.1038/s41467-022-34938-7. demonstrates sleep-like unsupervised replay as a mechanism to mitigate catastrophic forgetting in neural networks.

Tanha, J., N. Samadi, Y. Abdi, and N. Razzaghi-Asl. 2022. "Cpssds: Conformal Prediction for Semi-Supervised Classification on Data Streams." *Information Sciences* 584:212–234. https://doi.org/10.1016/j.ins.2021.10.068.

Teng, R., A. Wang, and H. Li. 2025. "Addressing Catastrophic Forgetting and Beyond: Key Challenges in Continual Learning." In *Proceedings of the 7th International Symposium on Advanced Technologies and Applications in the Internet of Things (ATAIT)*, 1–11. Kusatsu, Shiga, Japan. http://ceur-ws.org/Vol-4082/paper1.pdf,cCBY4.0.

Tripathi, R. K., L. Raja, and A. Kumar. 2021. "A Cluster Based Classification for Imbalanced Data Using Smote." In *IOP Conference Series: Materials Science and Engineering*, 012080. Jaipur, India. https://doi.org/10.1088/1757-899X/1099/1/012080.

Untoro, M. C., M. Praseptiawan, and M. Widianingsih. 2020. "Evaluation of Decision Tree, K-Nn, Naive Bayes and Svm with Mwmote on Uci Dataset." *Journal of Physics: Conference Series* 1477 (3): 032005. https://doi.org/10.1088/1742-6596/1477/3/032005.

Vanschoren, J., J. N. van Rijn, B. Bischl, and L. Torgo. 2014. "Openml: Networked Science in Machine Learning." *ACM SIGKDD Explorations Newsletter* 15 (2): 49–60. https://doi.org/10.1145/2641190.2641198.

Vito, S. 2008. "Air Quality." (UCI) Machine Learning Repository. https://doi.org/10.24432/C59K5F.

Wan, J. S. W., and S. D. Wang. 2021. "Concept Drift Detection Based on Pre-Clustering and Statistical Testing." *Journal of Internet Technology* 22 (2): 465–472.

Wang, L., X. Zhang, H. Su, et al. 2024. "A Comprehensive Survey of Continual Learning: Theory, Method and Application." *IEEE Transactions on Pattern Analysis & Machine Intelligence* 46 (8): 5362–5383. https://doi.org/10.1109/TPAMI.2024.3367329.

Wang, Y. 2019. "An Ensemble Learning Imbalanced Data Classification Method Based on Sample Combination Optimization." *Journal of Physics: Conference Series* 1284 (1): 012035. https://doi.org/10.1088/1742-6596/1284/1/012035.

Wang, Z., E. Yang, L. Shen, and H. Huang. 2025. "A Comprehensive Survey of Forgetting in Deep Learning Beyond Continual Learning." *IEEE Transactions on Pattern Analysis & Machine Intelligence* 47 (3): 1464–1483. https://doi.org/10.1109/TPAMI.2024.3498346.

Wen, W., F. Yan, Y. Chen, and H. Li. 2020. "Autogrow: Automatic Layer Growing in Deep Convolutional Networks." In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2020)*, 833–841. Virtual Conference: Association for Computing Machinery. https://doi.org/10.1145/3394486.3403126.

Wen Wang, L., Guo, W., and Cheng Yan Y. 2021. "An Online Weighted Sequential Extreme Learning Machine for Class Imbalanced Data Streams." *Journal of Physics: Conference Series* 012008. https://doi.org/10.1088/1742-6596/1994/1/012008.

Werner, E., N. Kumar, M. Lieber, S. Torge, S. Gumhold, and W. E. Nagel. 2024. "Towards Computational Performance Engineering for Unsupervised Concept Drift Detection – Complexities, Benchmarking, Performance Analysis." In *Proceedings of the 13th International Conference on Data Science, Technology and Applications (2024)*, edited by A. Cuzzocrea, E. Benkhelifa, O. Gusikhin, and S. Hammoudi, 318–329. VOl. 1. Dijon, France: SciTePress. https://doi.org/10.5220/0012758600003756.

Witten, I. H., E. Frank, and M. A. Hall. 2016. *The WEKA Workbench: Online Appendix for "Data Mining: Practical Machine Learning Tools and Techniques"*. 4th ed. Cambridge, MA: Morgan Kaufmann.

Wu, Y., Y. Chen, and L. Wang. 2019. "Large Scale Incremental Learning." Conference on Computer Vision and Pattern Recognition (CVPR), 836–845. Long Beach, CA. CVPR 2019/html/Wu Large Scale Incremental Learning CVPR 2019 paper.html, bias Correction (BiC) introduced within this work. https://doi.org/10.1109/CVPR.2019.00091.

Xiang, Q., L. Zi, X. Cong, and Y. Wang. 2023. "Concept Drift Adaptation Methods Under the Deep Learning Framework: A Literature Review." *Applied Sciences* 13 (11): 6515. https://doi.org/10.3390/app13116515.

Xuan, J., J. Lu, and G. Zhang. 2020. "Bayesian Nonparametric Unsupervised Concept Drift Detection for Data Stream Mining." *ACM Transactions on Intelligent Systems and Technology* 12 (1): 5:1–5:22. https://doi.org/10.1145/3420034.

Ye, S., B. Kim, and B. Jeon. 2025. "Magnitude Estimation of Multiple Data Drifts for Image Classification Neural Network." 2025 19th International Conference on Ubiquitous Information Management and Communication (IMCOM), 1–4. Bangkok, Thailand. https://doi.org/10.1109/IMCOM64595.2025.10857512.

Yin, W., C. Xie, and B. Chen. 2023. "Forward Explanation: Why Catastrophic For- Getting Occurs." International Conference on Learning Representations (ICLR) 2024 Workshop Track. Vienna, Austria. https://openreview.net/forum?id=ZyMXxpBfct

Yoon, J., E. Yang, J. Lee, et al. 2018. *Lifelong Learning with Dynamically Expandable Networks*. Vancouver, BC: OpenReview.net. https://doi.org/10.48550/arXiv.1708.01547.

Zenke, F., B. Poole, and S. Ganguli. 2017. "Continual Learning Through Synaptic Intelligence." *Proceedings of the 34th International Conference on Machine Learning (ICML)* 70: 3987–3995. http://proceedings.mlr.press/v70/zenke17a.html.

Zhang, L., B. Tan, T. Liu, et al. 2019. "Classification Study for the Imbalanced Data Based on Biased-Svm and the Modified Over-Sampling Algorithm." *Journal of Physics: Conference Series* 1237 (32): 022052. https://doi.org/10.1088/1742-6596/1237/2/022052.

Zhang, Z., J. Qiu, and W. Dai. 2019. "A New Improved Boosting for Imbalanced Data Classification." *IOP Conference Series: Materials Science and Engineering*: 012047. https://doi.org/10.1088/1757-899X/533/1/012047.

Zheng, S. 2019. "Labelless Concept Drift Detection and Explanation." https://research.tue.nl/en/studentTheses/labelless-concept-drift-detection-and-explanation/.

Zhong, Y., J. Zhou, P. Li, and J. Gong. 2023. "Dynamically Evolving Deep Neural Networks with Continuous Online Learning." *Information Sciences* 646:119411. https://doi.org/10.1016/j.ins.2023.119411.